# FARSHORE

## The Odyssey: Arcade Trials

Phase 2: Interface structure and AI interaction design



Bibek Pathak

CMPS 4200 HCI / October 8, 2026

# The five arcade trials

Each shore uses different controls and shows an active rival

| Shore | Player interaction | Visible rival action |
| --- | --- | --- |
| Cyclops | Grid stealth | Blind smash from noise clues |
| Circe | Falling-potion match-3 | Own board, chains and curses |
| Sirens | Four-lane melody memory | Composes and targets missed lanes |
| Scylla & Charybdis | Shield paddle and rowing | Six heads with pulsing vortex |
| Ithaca | Turn-based archery | Own scored shot or ring block |

# Players and design needs

Phase 1 persona sketches guide the flows

## Maya, 21

Win through skill and understand why the rival behaves differently.

Fixed rules and visible rival warnings.

## Daniel, 19

Experience the Odyssey without a confusing first encounter.

Short practice and one clear objective per shore.

## Rosa, 34

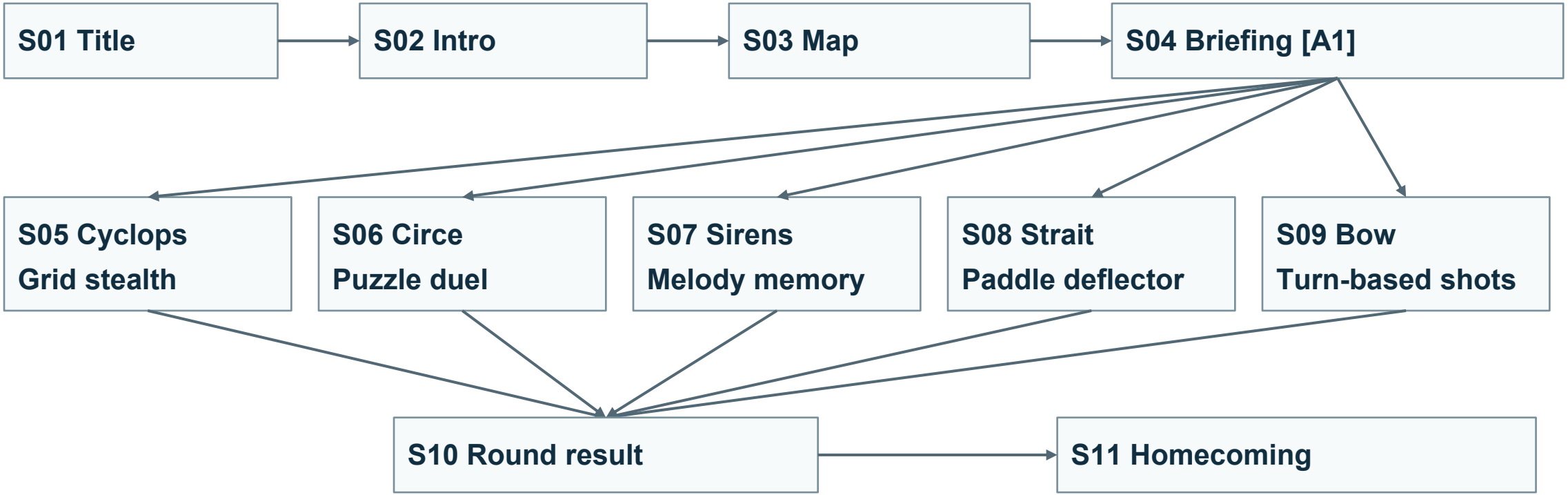
Know what the model changes and keep control over its result.

Model labels, strategy review and a usable fallback.

Provisional design references. Interview evidence has not been supplied.

# Screen hierarchy

A1 in the briefing is the only model feature in arcade mode

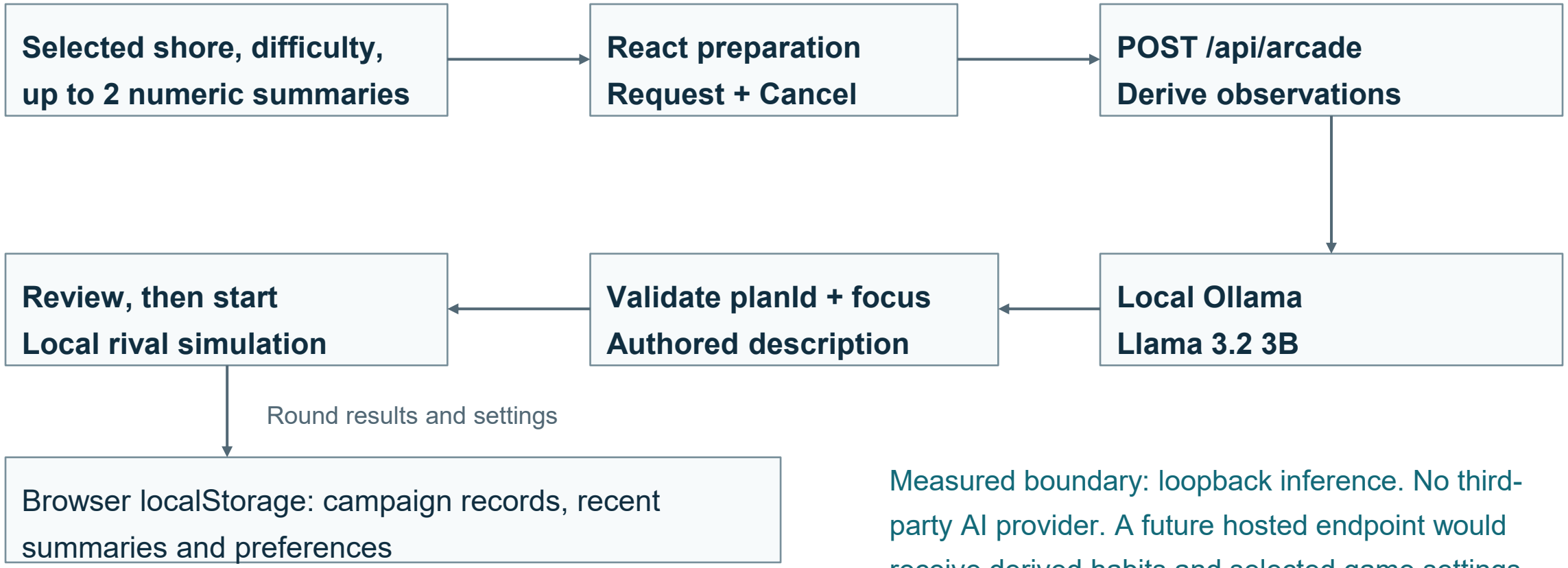



Next trial opens its briefing after a clear or assist. The fifth passage opens Homecoming.

Shared overlays: S12 Settings, S13 Pause/leave, S14 Restart, S15 Assisted passage

# Model data flow and privacy

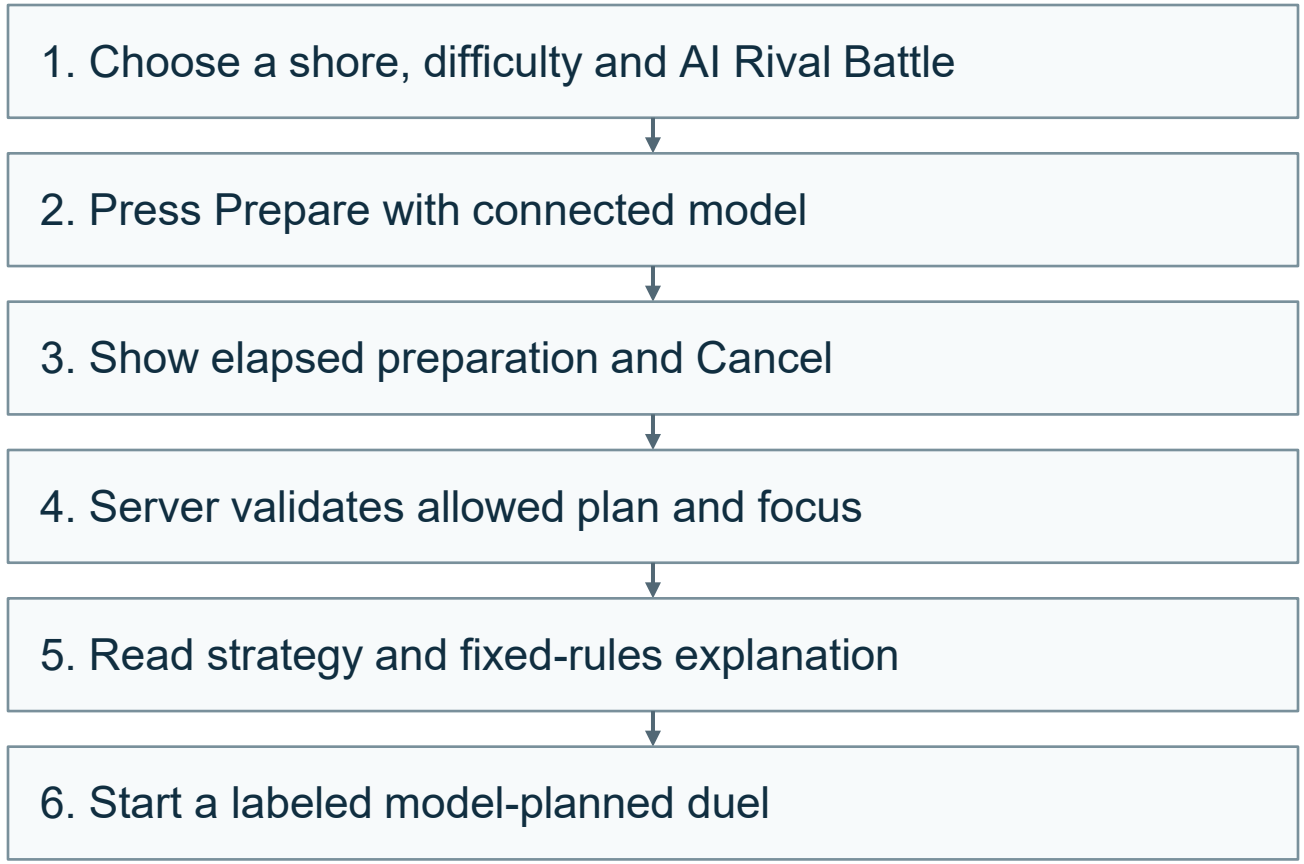
Numeric summaries prepare a strategy before the round




Measured boundary: loopback inference. No third-party AI provider. A future hosted endpoint would receive derived habits and selected game settings.

# F2: Strategy preparation and review

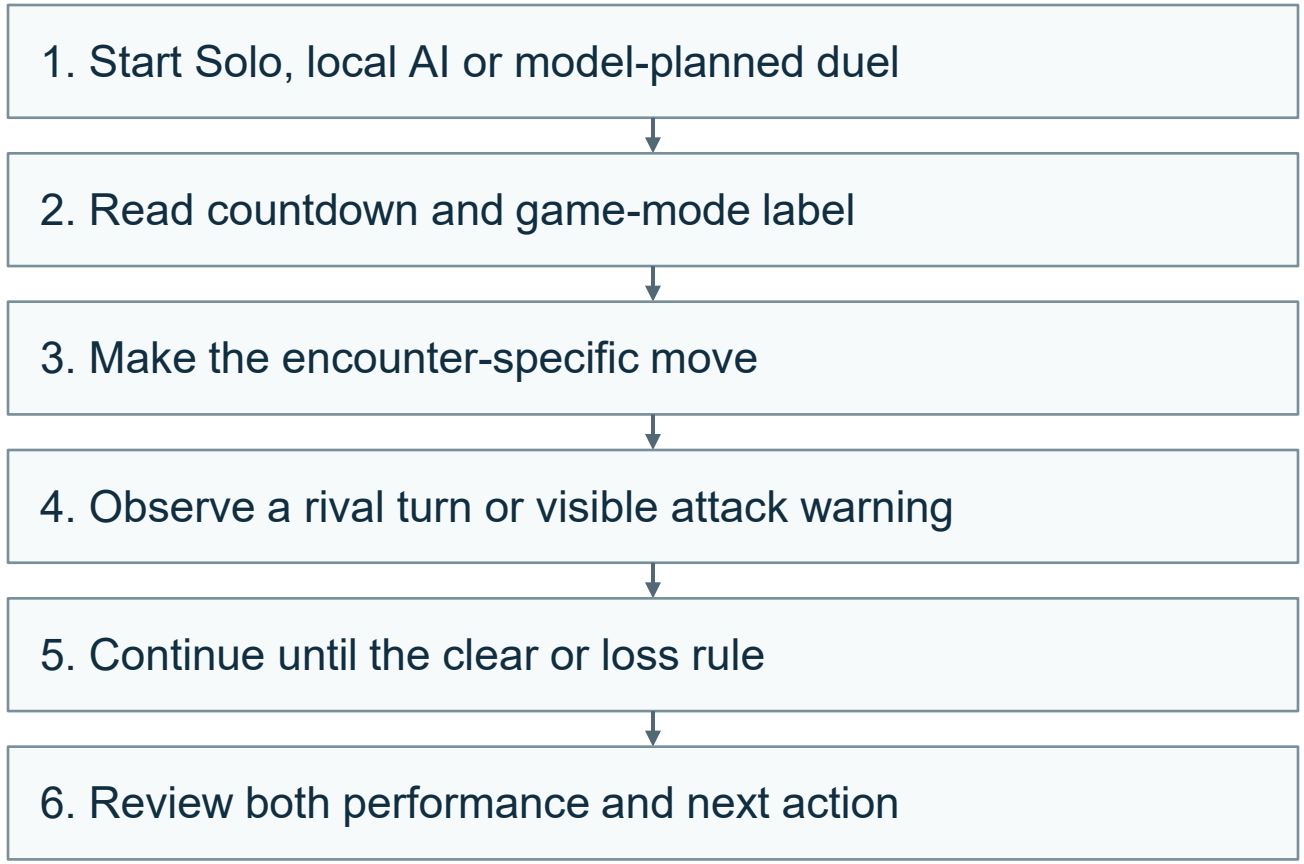
Maya and Rosa / player control before a match




- Cancel preserves progress.
- Prepare another replaces the candidate.
- Edit difficulty or choose Solo to discard.
- Proposed: a dedicated Discard action.

# F3: Active play and recovery

Maya and Daniel / local simulation keeps play responsive




- Warnings show what the rival will do.
- Pause stops clocks and held controls.
- Loss offers retry or short practice.
- Assistance has an honest report label.

# AI preparation states

Initial, loading and reviewed success

## Initial

- Difficulty and mode
- Local rival ready
- Prepare with model

## Loading

- Preparing strategy
- 12 seconds elapsed
- Cancel / Practice

## Success

- Chain Alchemist
- Fixed Normal rules
- Start / Prepare another

W05-W07 / F2. Preparation consumes no gameplay time.

# F4: Uncertainty and error recovery

Rosa and Maya / no dead end and no invented confidence score

## Uncertain or unsupported

- Unsupported strategy
- Nothing changed
- Retry / local AI / Solo

## Service error

- Unavailable or timed out
- Your voyage is kept
- Retry / check / local play

W08-W09 / F4. Invalid input is rejected before model contact.

# AI technical spike

Six synthetic use-case inputs, real local Llama 3.2 3B calls

## Min 7.317s  Mean 8.279s  Max 11.613s

| Sample | Seconds | Selected plan | Review |
|---|---|---|---|
| S1 First-time Cyclops | 11.613 | focused/center | Legal, advisory mismatch |
| S2 Circe after two losses | 7.597 | chain/balanced | Matches loss preference |
| S3 Sirens with red-lane misses | 7.317 | echo/balanced | Legal, lane targeting is local |
| S4 Strait after one hull loss | 7.889 | pincer/balanced | Legal, advisory mismatch |
| S5 Archery after low ring score | 7.614 | feint/balanced | Legal block-shot policy |
| S6 Cyclops with left-side habit | 7.641 | focused/left | Schema fixes left focus |

Six legal outputs from one sequential run. Tactical quality still needs evaluation.

# Failure evidence and design decisions

Observed HTTP responses and designed UI recovery

| Triggered condition | Observed response | Designed recovery |
| --- | --- | --- |
| Nonexistent provider model | HTTP 404, model not found | Check / retry / local play |
| Malformed provider JSON | HTTP 400, unexpected EOF | Explain failure / retain selections |
| Unsupported difficulty | App HTTP 400 before inference | Supported-choice guidance |

**Wait design: elapsed time and Cancel. No round runs during preparation.**

No live 429 or capacity benchmark. Local API calls had no billed fee.

Hardware, electricity and future hosting costs are unmeasured.

# Phase 2 design and next evaluation

Implemented behavior and proposed improvements

## Implemented

- Five distinct playable trials
- Model review, cancellation and regeneration
- Local saves and warning, practice and assistance

## Proposed or awaiting evidence

- Explicit Discard and provider privacy detail
- Interview-backed persona validation
- Actual peer critique and fairness playtests

Review appendix: all 15 screens, 5 flows, 28 wireframes and exact failure bodies.

# Content strategy

Review appendix / screens S01-S05

| Screen | Displayed information | Model and storage |
|---|---|---|
| S01 Title | Begin/resume, five-trial premise, progress count and Settings. | No model call. Read browser-local save. |
| S02 Introduction | Odysseus and the voyage, next/skip. | Authored story. Skip preserves the new voyage. |
| S03 Voyage map | Five shores, cleared/assisted/pending states, locked stops, continue. | No model call. Every earlier shore must be passed. |
| S04 Encounter briefing [A1] | Rival, myth context, objective, controls, difficulty, mode, practice and preparation. | Only pretrained-model integration point: A1 optional rival planning. |
| S05 Cyclops round | Grid, sheep, rocks, exits, noise and fixed smash warning, move controls. | Local opponent engine. One evade per warning. |

# Content strategy

Review appendix / screens S06-S10

| Screen | Displayed information | Model and storage |
| --- | --- | --- |
| S06 Circe round | Two puzzle boards, both next pieces, curses and scores, rotate/drop. | Circe plays independently. No model request during a round. |
| S07 Sirens round | Four numbered/color lanes, Listen/Reply, sequence, Temptation. | Local melody logic targets missed lanes. Optional distinct sound. |
| S08 Strait round | Shield, six heads, warning paths, rowing control, vortex, hull and time. | Local coordinated attacks. Hold/release input is explicit. |
| S09 Archery round | 12 rings, wind, path preview, three controls, turn and both scores. | Player and rival use the same shot physics. Rival turn disables shooting. |
| S10 Round result | Clear/loss/practice, score, opponent score where relevant, retry and next. | Completed non-practice rounds update attempts. Failed replays retain past clears. |

# Content strategy

Review appendix / screens S11-S15

| Screen | Displayed information | Model and storage |
| --- | --- | --- |
| S11 Homecoming | Family, report of each clear or assisted passage, replay and restart. | No generated ending. Honest saved progress report. |
| S12 Settings overlay | Sound, still scenery, model connection check, earlier modes. | GET checks availability, not a generated rival strategy. |
| S13 Pause / leave overlay | Resume, leave warning and return to briefing. | Pause simulation and release held controls. Abandoned round gives no clear. |
| S14 New voyage confirmation | Explain replacement of current progress, keep or restart. | Replace save only after explicit confirmation. |
| S15 Assisted passage confirmation | Explain Assisted report label, keep trying or continue with help. | Authored progression choice. Replay can later earn a clear. |

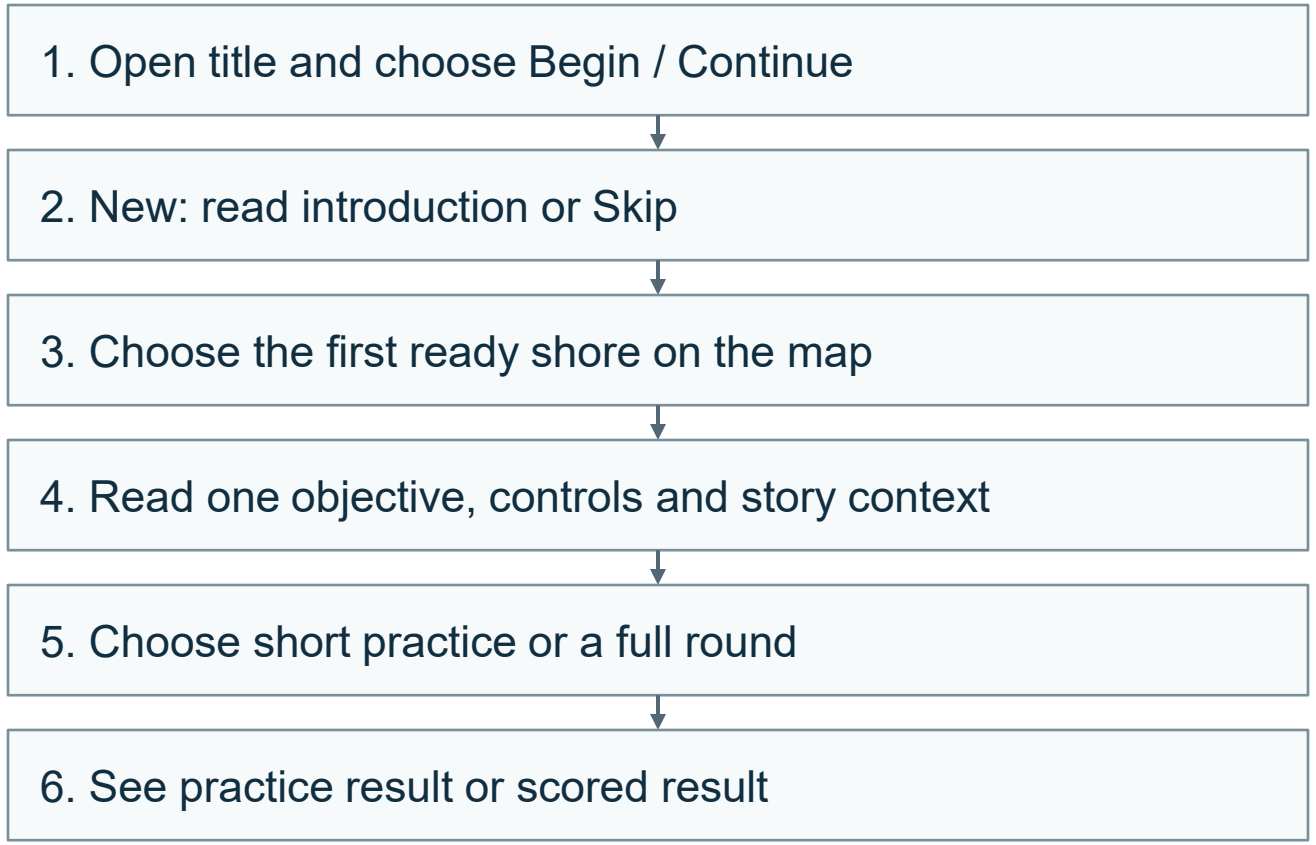
# A1 model contract

Review appendix / optional planning before a match

| | |
|---|---|
| **Input** | The browser sends a shore, difficulty and up to two numeric attempts. The server sends derived habits, required focus and win count to Llama. |
| **Output** | Llama chooses one of two strategy IDs. The game fixes the required focus. Strict validation produces an authored description for review. |
| **Timing and provider** | Six-call range: 7.317-11.613 s, mean 8.279 s. Local loopback Ollama. Server timeout 90 s, browser limit 95 s. No gameplay clock runs. |
| **Current data limit** | New games save zero for legacy early/late and lane durations. S6 injects left-lane history. Siren targeting uses current-round misses. |

# F1: Start, resume and practice

P2 Daniel / review appendix




## Existing progress

Continue retains clears and preferences. Restart requires S14 confirmation.

## Controls are unfamiliar

Practice leaves campaign progress unchanged. Return to the same briefing.

Screens: S01, S02, S03, S04, S05-S09, S10

# F4: Failed-plan recovery

P3 Rosa and P1 Maya / review appendix

1. Observe service error or rejected strategy
2. Read a specific explanation with progress kept
3. Review selected shore, difficulty and connection
4. Choose retry, changed input, Solo or local AI
5. Review a fresh valid plan or local label
6. Start only after an explicit user choice

Screens: S04 / A1 uncertain, error, invalid input; S12

## Poor / empty / unsupported output

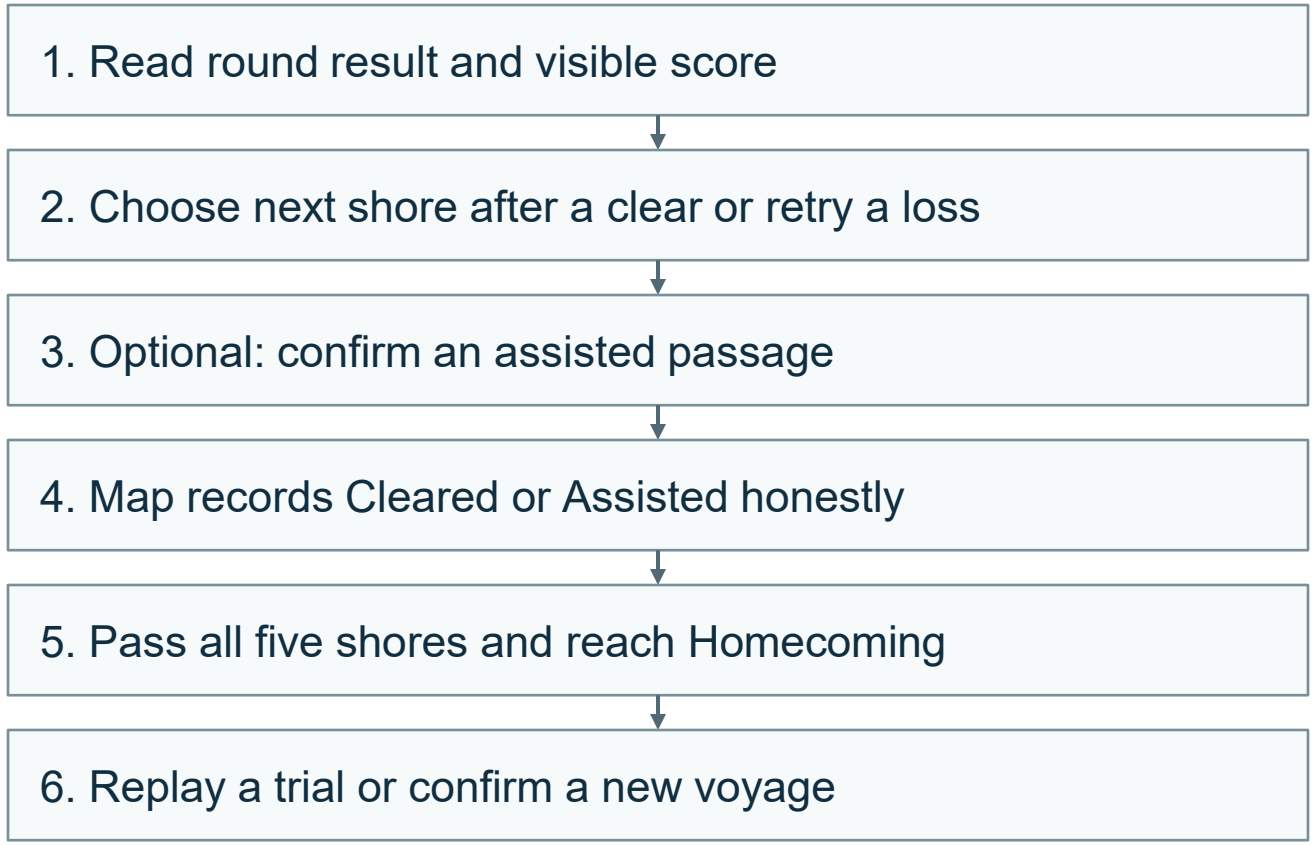
Invalid JSON, extra fields, illegal plan or wrong focus is rejected. No match starts. For a legal but unwanted tactic, regenerate or reject.

## Offline / timeout / rate limit

Retry when ready, cancel a long wait, or choose Solo/local AI. Provider 429 maps to a busy-service message.

# F5: Progress and homecoming

P2 Daniel, P1 Maya and P3 Rosa / review appendix




## Reconsider help

Keep trying closes S15 without changing progress. A later successful replay upgrades Assisted to Cleared.

## New voyage

S14 explains that restarting replaces progress. Keep voyage returns unchanged.

Screens: S10, S15, S03, S11, S14

# Edge cases and recovery

Review appendix / EC1-EC3

| Scenario and states | Recovery | Evidence or design status |
| --- | --- | --- |
| EC1 Unavailable / timeout / 429<br>F4, A1 error | Keep progress. Retry, Cancel, Solo or a labeled local AI duel. Preparation never consumes game time. | E1 observes real provider 404. 90s timeout and 429 mapping exist; those cases were not induced live. |
| EC2 Poor, wrong or empty output<br>F2/F4, A1 uncertain | Reject invalid JSON or unsupported plan/focus. Regenerate or change inputs. Legal but unwanted tactics remain rejectable. | Strict schema and server validation; fresh spike measures legal outputs, not tactical enjoyment. |
| EC3 Invalid / unsupported input<br>F4, S04 validation | Explain supported choices and keep valid settings. Reject invalid input before model contact. Inline field guidance is proposed. | E3 records actual application 400 for impossible difficulty. |

# Edge cases and recovery

Review appendix / EC4-EC6

| Scenario and states | Recovery | Evidence or design status |
| --- | --- | --- |
| EC4 Cancel or late reply<br>F2, A1 loading | Abort and increase request generation ID. Ignore old replies after cancel, input edit, navigation or regeneration. | Implemented request token and AbortController. No clear or score can arrive from a model reply. |
| EC5 Inappropriate / biased output<br>F2/F4, A1 uncertain | Allow permitted strategy IDs. Show authored descriptions. Reject generated prose and all extra fields. | This limits the output surface; fair tactical behavior still needs player evaluation. |
| EC6 Hidden tab / lost held input<br>F3, S13 pause | Pause simulation and release rowing/paddle controls. Resume explicitly. Explain that reload restarts the round. | Implemented focus/visibility handling. Completed progress stays local. |

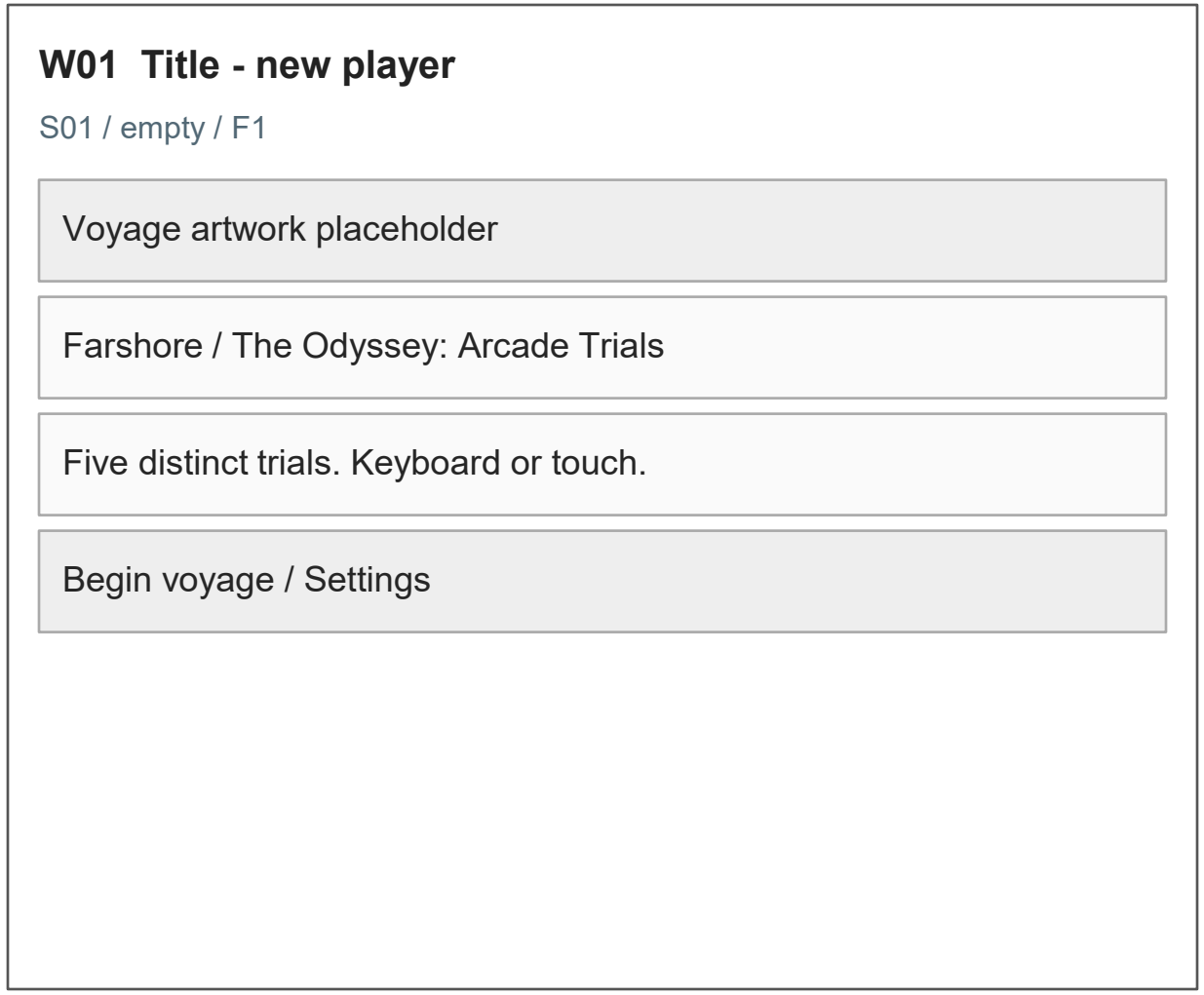
# Edge cases and recovery

Review appendix / EC7-EC8

| Scenario and states | Recovery | Evidence or design status |
| --- | --- | --- |
| EC7 Local storage unavailable<br>F1/F5, S01/S03 | Show the implemented persistent saving warning. Keep the in-memory voyage and explain that reload may lose progress. | Storage exceptions trigger a persistent status notice. This behavior is implemented. |
| EC8 Migrated old save / abandoned round<br>F1/F5, S03/S10 | Retain old clears, insert Circe/strait pending. Clear earlier shores before replaying later ones. No clear for an unfinished round. | Save schema 2 migration. Assisted and practice labels remain explicit. |

# W01: Title - new player

Review appendix / low-fidelity screen and state




## Implemented

A single primary action starts the game without AI setup.

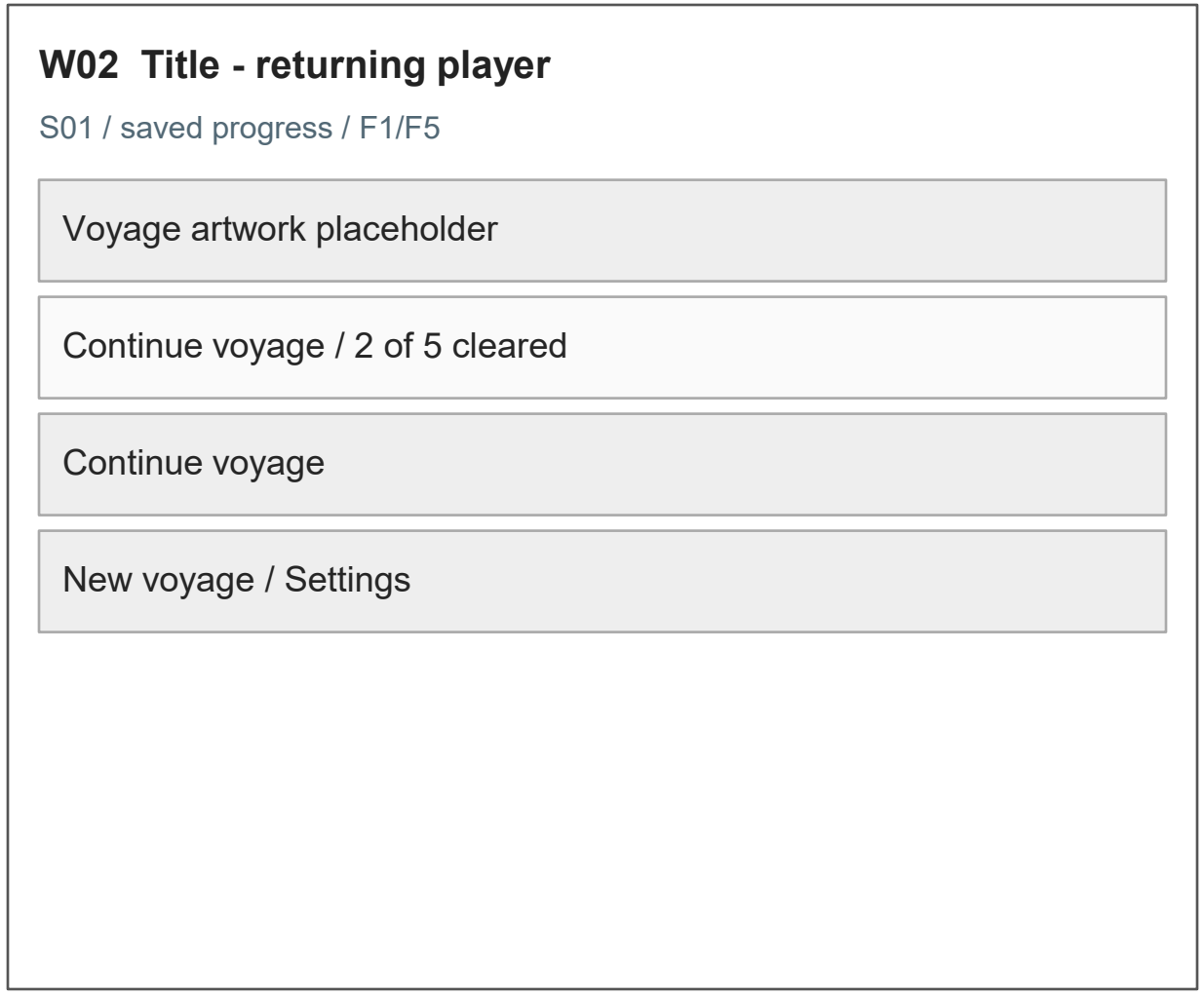
Screen S01

Flow F1

State empty

# W02: Title - returning player

Review appendix / low-fidelity screen and state




## Implemented

Continue reads browser-local progress.

New voyage opens S14.

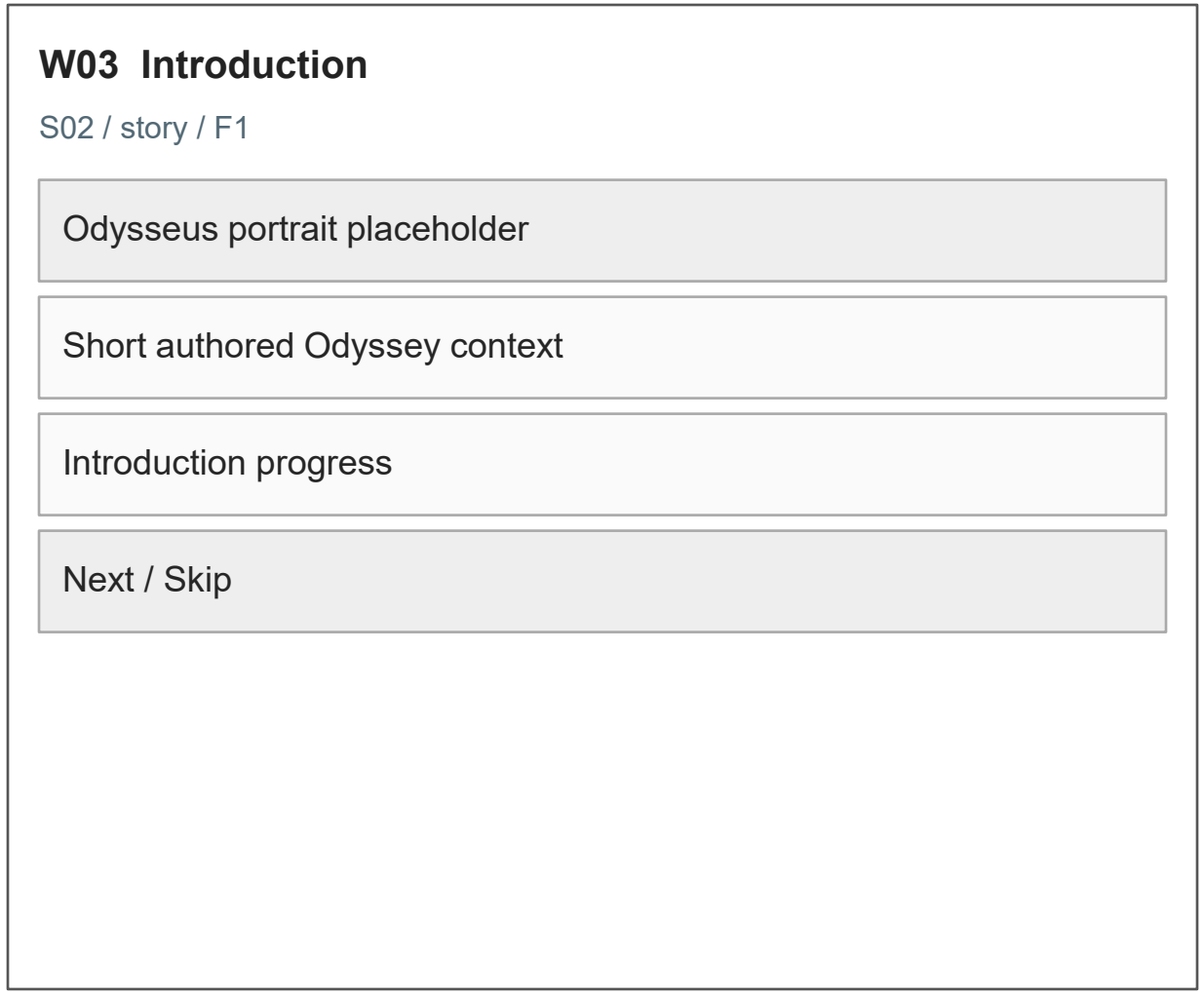
Screen S01

Flow F1/F5

State saved progress

# W03: Introduction

Review appendix / low-fidelity screen and state




## Implemented

Skip moves to the map. This story text is authored, with no generation.

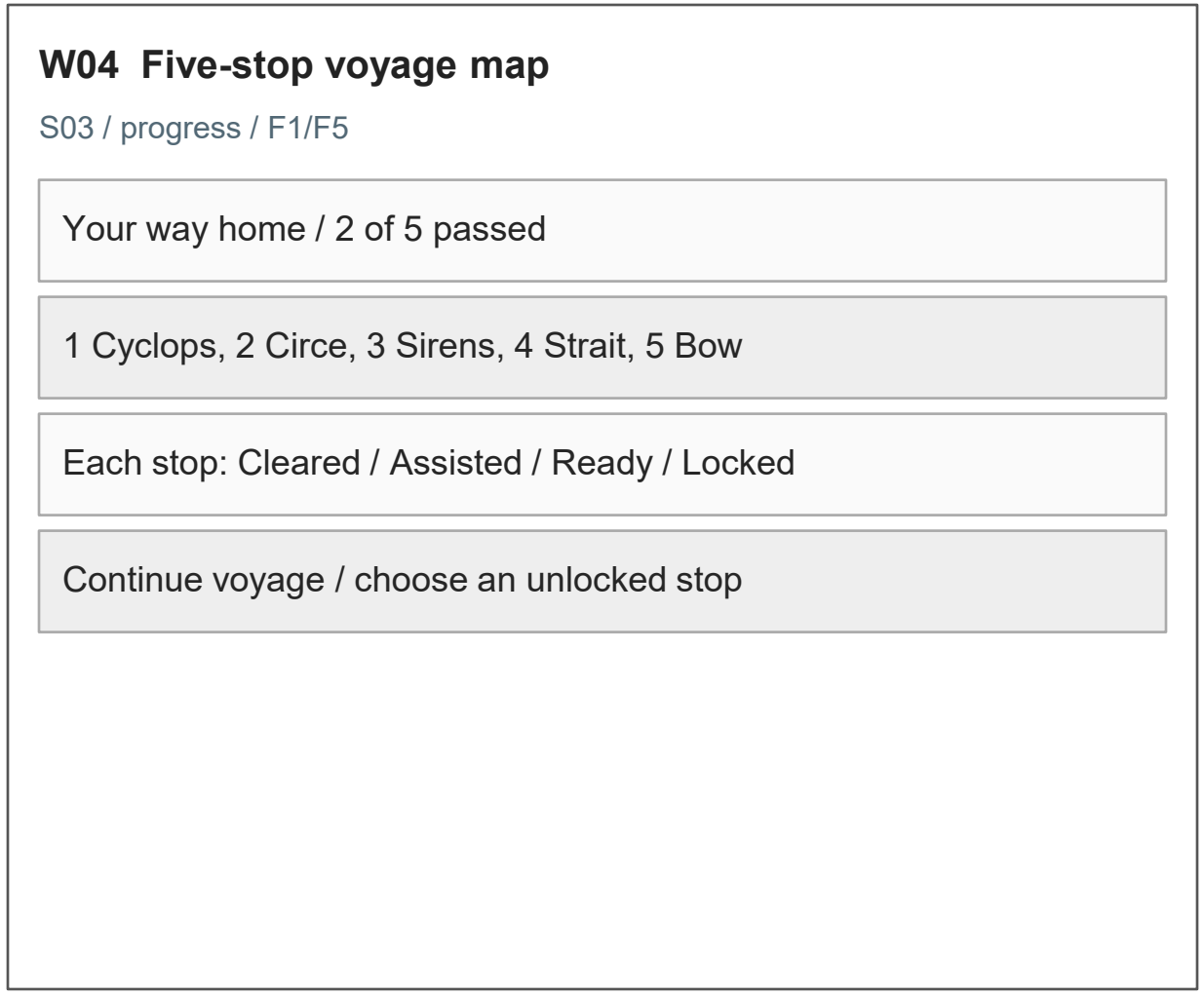
Screen S02

Flow F1

State story

# W04: Five-stop voyage map

Review appendix / low-fidelity screen and state




## Implemented

Locked later trials explain the preceding requirement. Migration adds two new pending shores.

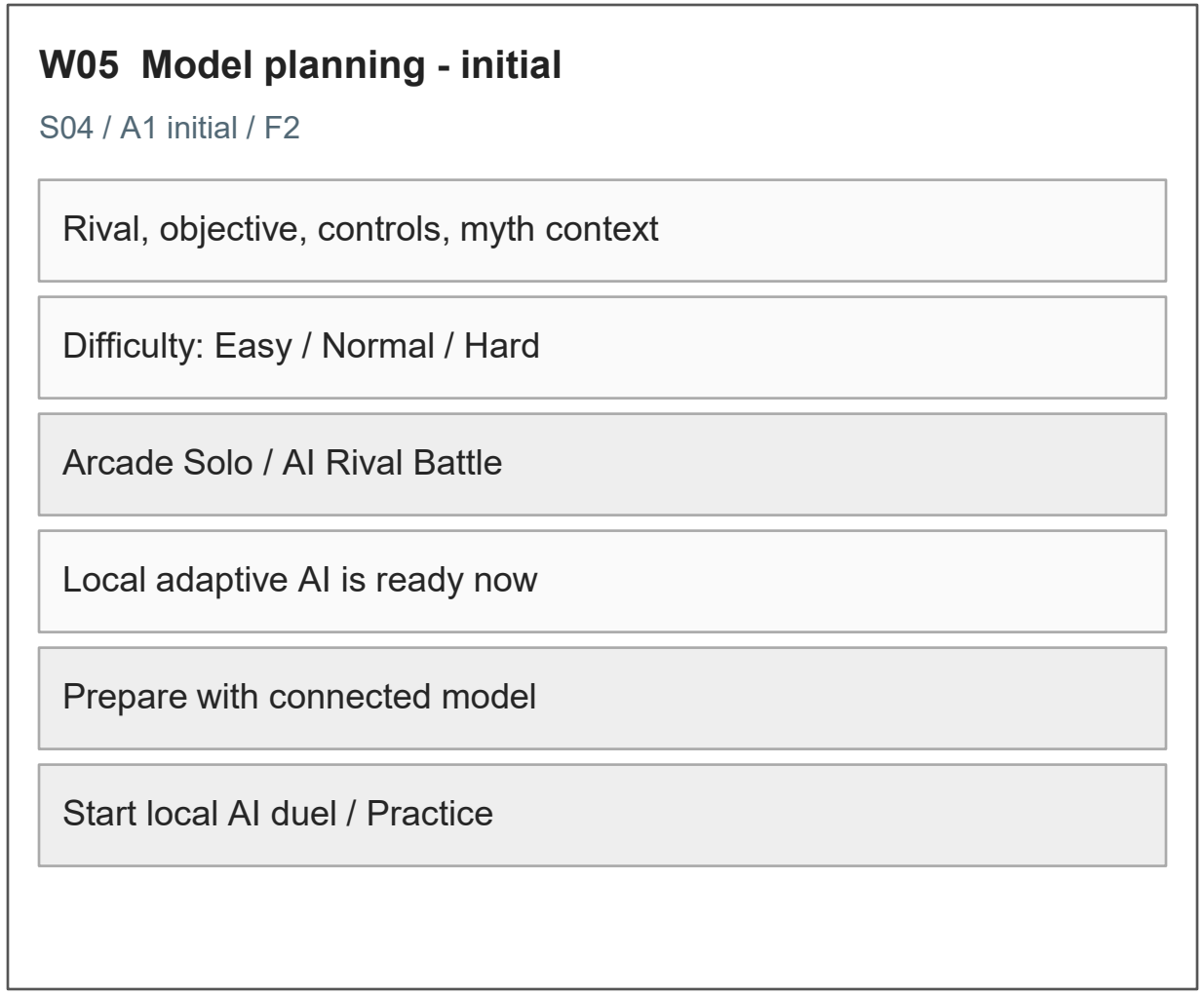
Screen S03

Flow F1/F5

State progress

# W05: Model planning - initial

Review appendix / low-fidelity screen and state




**Implemented; privacy detail Proposed**

A1 begins only on Prepare. Numeric attempt summaries provide context. Proposed detail: visible loopback-provider disclosure.

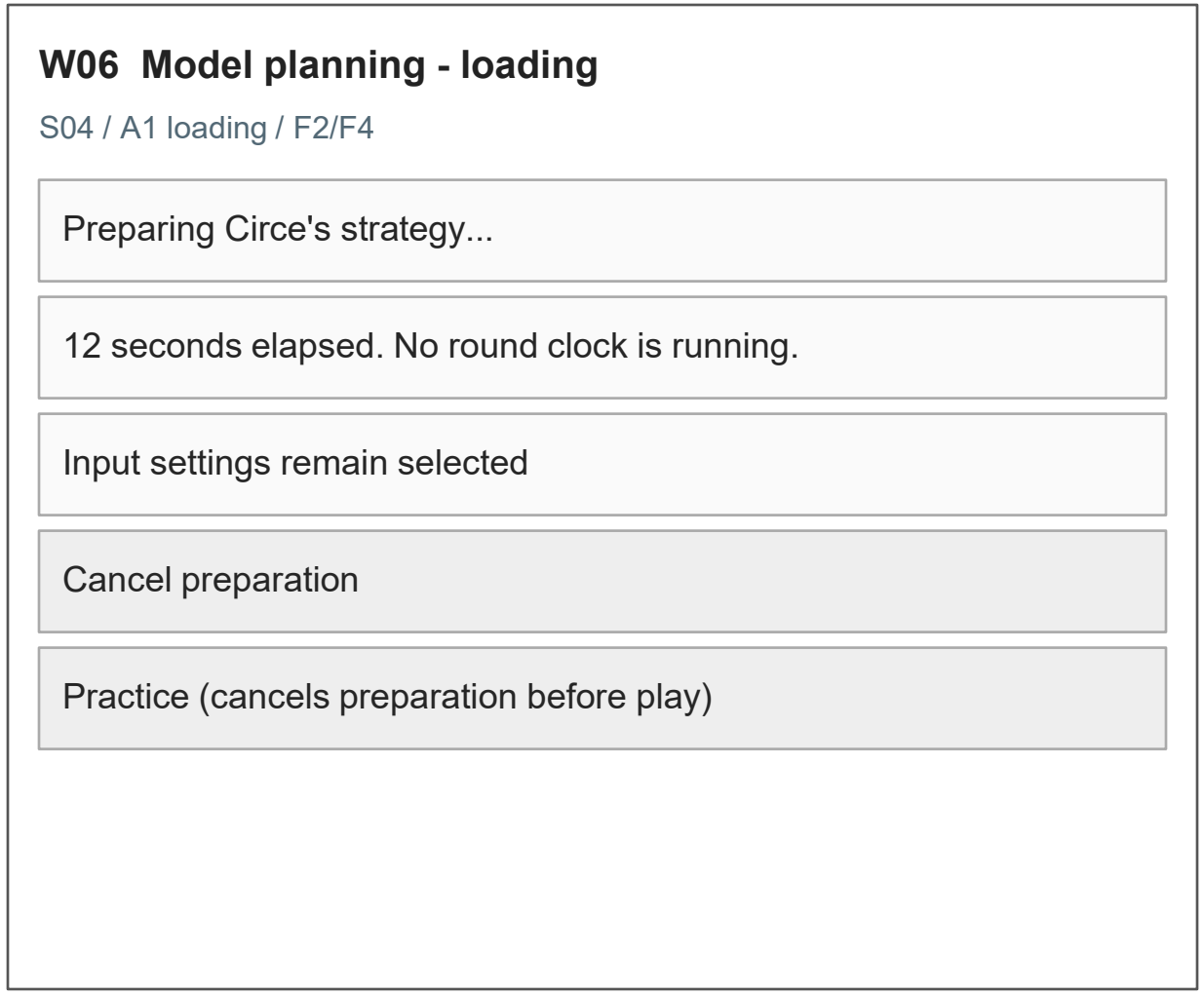
Screen S04

Flow F2

State A1 initial

# W06: Model planning - loading

Review appendix / low-fidelity screen and state




## Implemented

Clear feedback appears immediately. Cancel ignores stale replies. Start is disabled while loading.

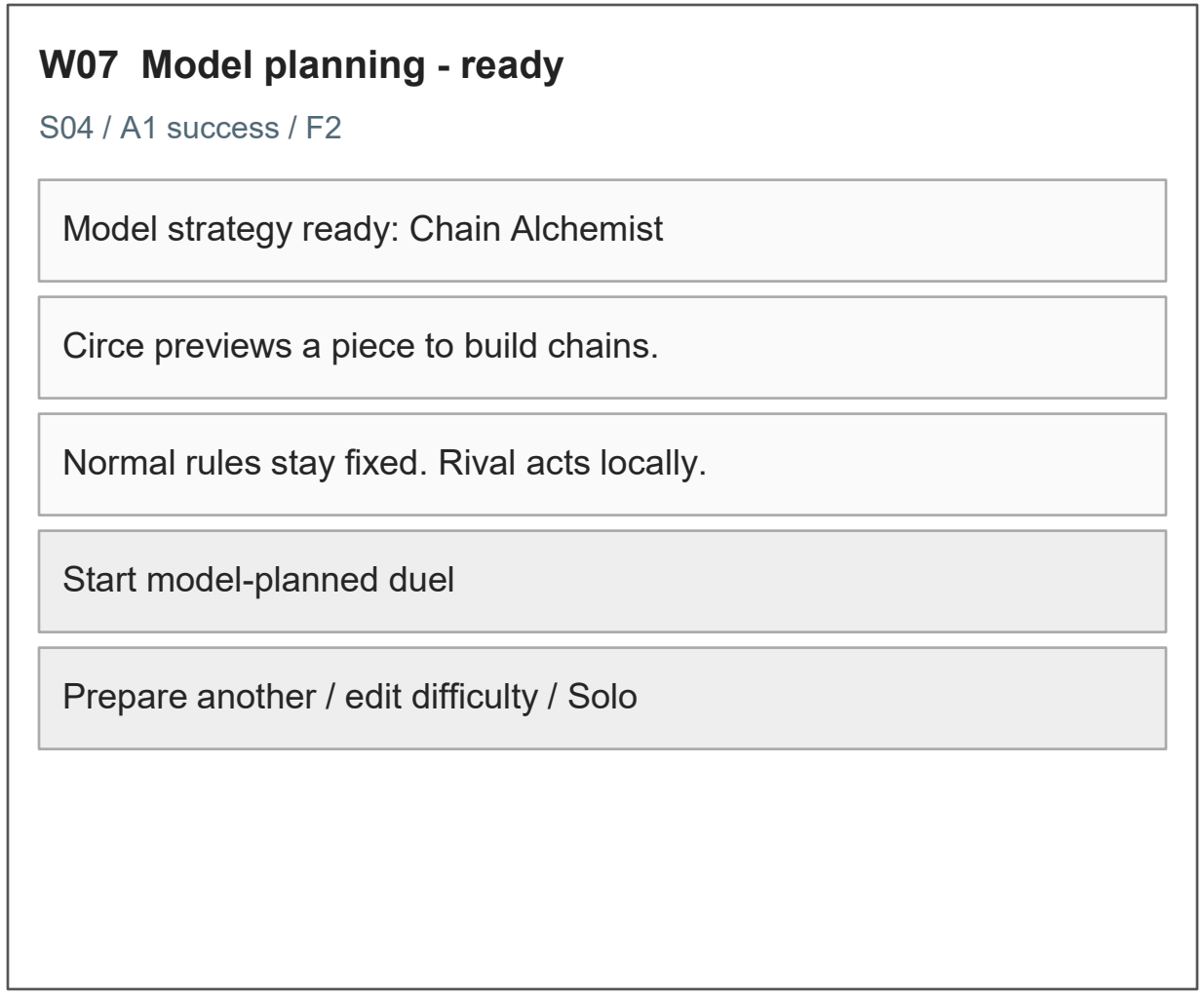
Screen S04

Flow F2/F4

State A1 loading

# W07: Model planning - ready

Review appendix / low-fidelity screen and state




## Implemented

Review precedes play. Changing mode or difficulty discards the ready plan. No scoring changes during preparation.

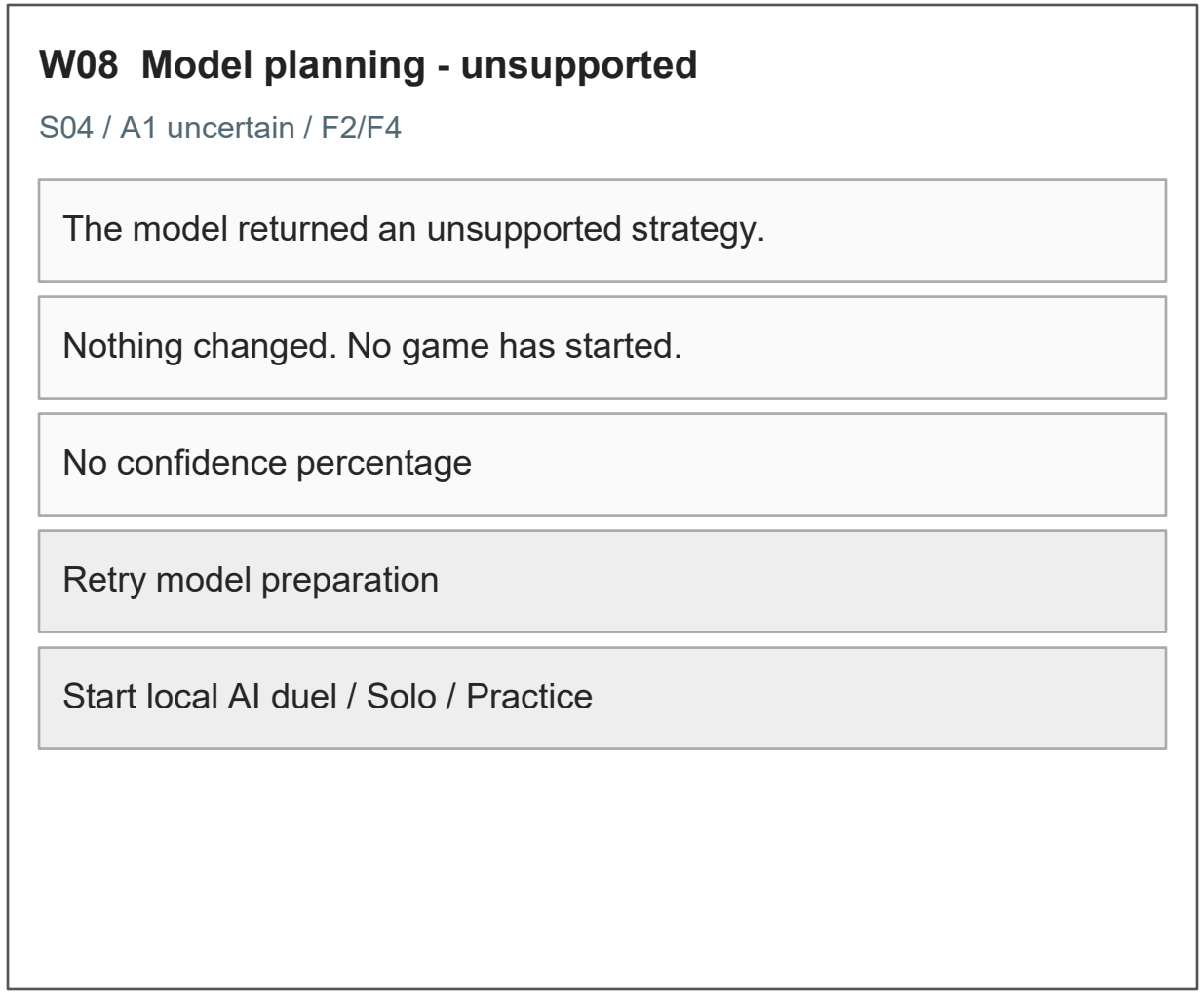
Screen S04

Flow F2

State A1 success

# W08: Model planning - unsupported

Review appendix / low-fidelity screen and state




## Implemented

Invalid or extra fields cannot become a rival plan. Uncertain describes suitability, not a measured probability.

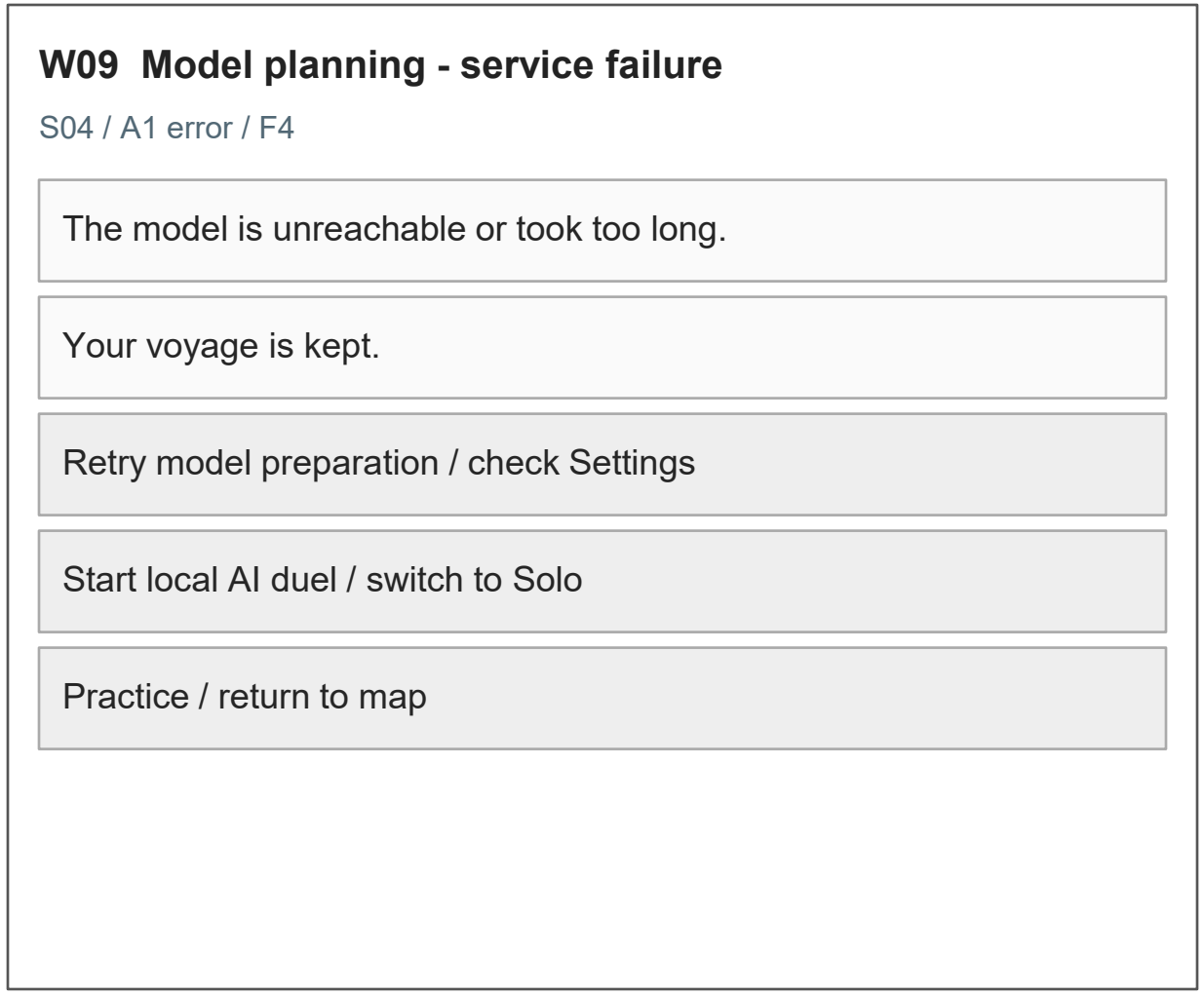
Screen S04

Flow F2/F4

State A1 uncertain

# W09: Model planning - service failure

Review appendix / low-fidelity screen and state




**Implemented**

A failed request never silently becomes model-planned play. Every fallback has its own label.

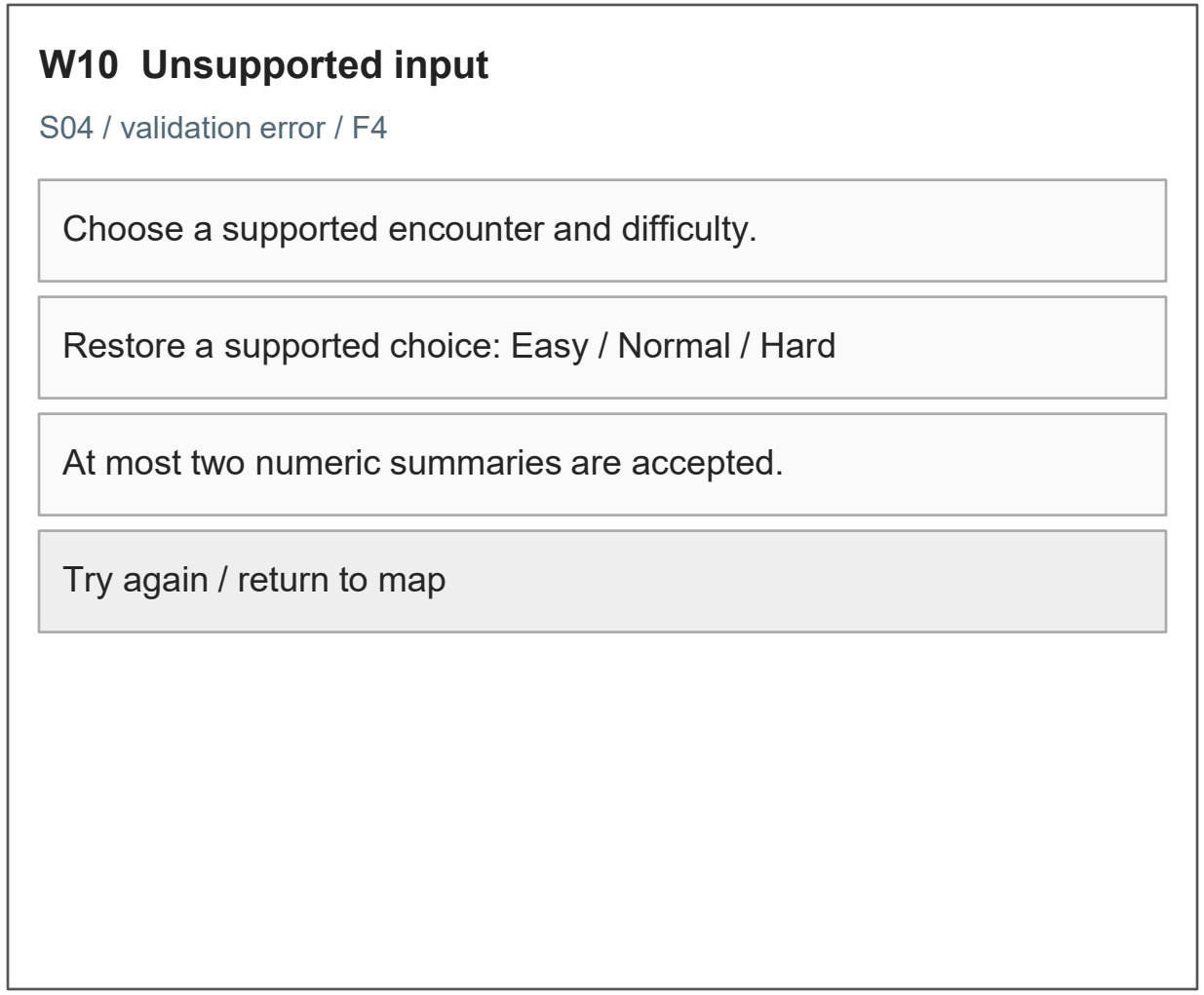
Screen S04

Flow F4

State A1 error

# W10: Unsupported input

Review appendix / low-fidelity screen and state




**Server validation implemented; inline state Proposed**

The visible selectors constrain ordinary input. The server rejects malformed or tampered requests before inference.

Screen S04

Flow F4

State validation error

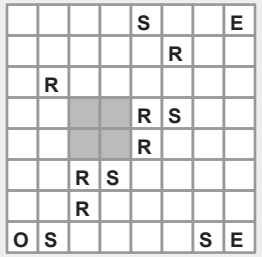
# W11: Polyphemus grid

Review appendix / low-fidelity screen and state

### W11 Polyphemus grid

S05 / player / smash warning / F3

Grid stealth / turn / hits / current rival mode




O = Odysseus
S = sheep / R = rock
E = exit
Gray = warned smash

Marked 2 x 2 smash area. One evade left.

Up / Left / Right / Down / Sneak-Dash / Hide

Pause

## Implemented

Blind rival uses noise clues. The visible target locks before the single evade. Exits and sheep have symbols.

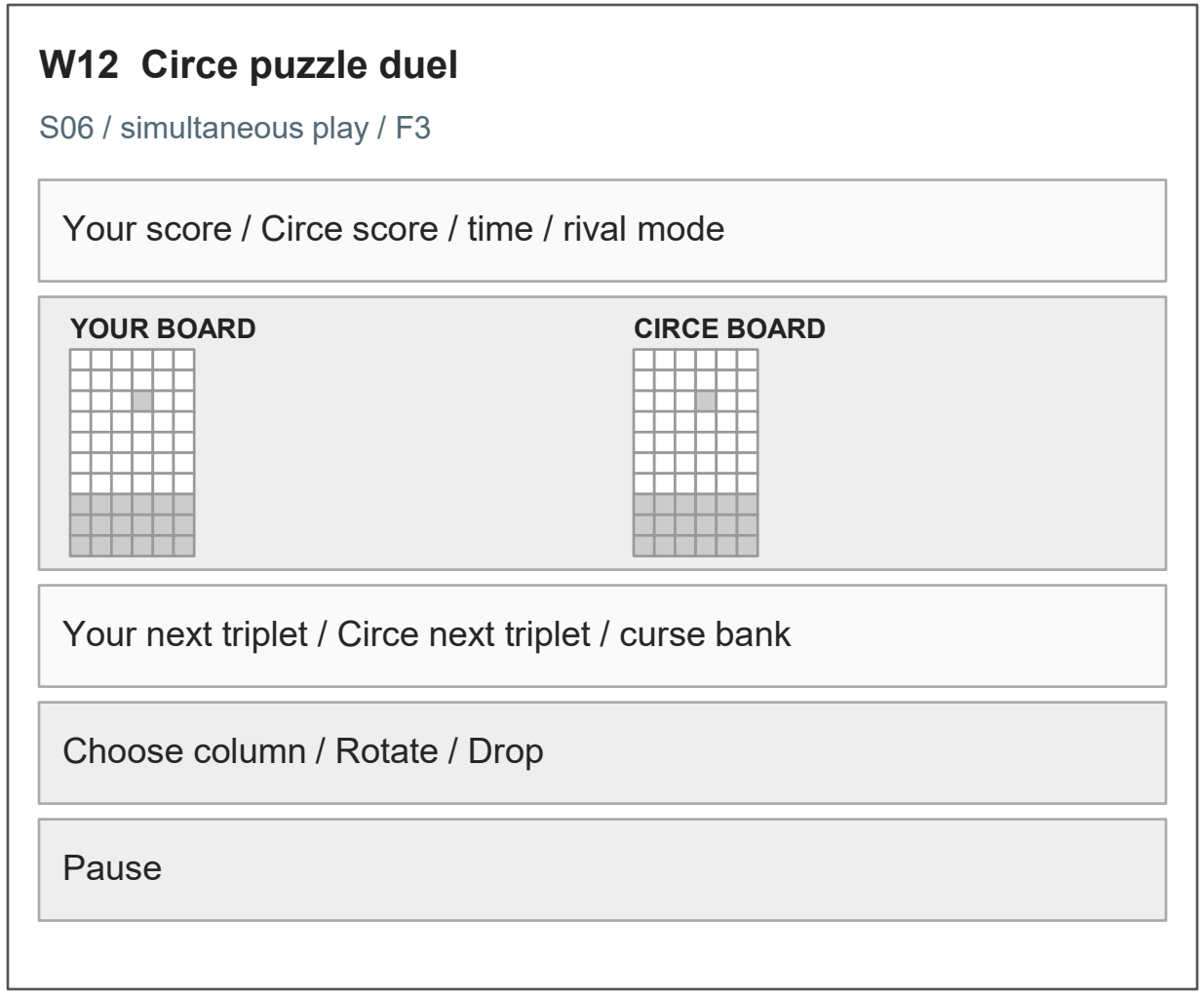
Screen S05

Flow F3

State player / smash warning

# W12: Circe puzzle duel

Review appendix / low-fidelity screen and state




## Implemented

Two boards make the opponent's moves visible. Moly matches purify swine and send curses.

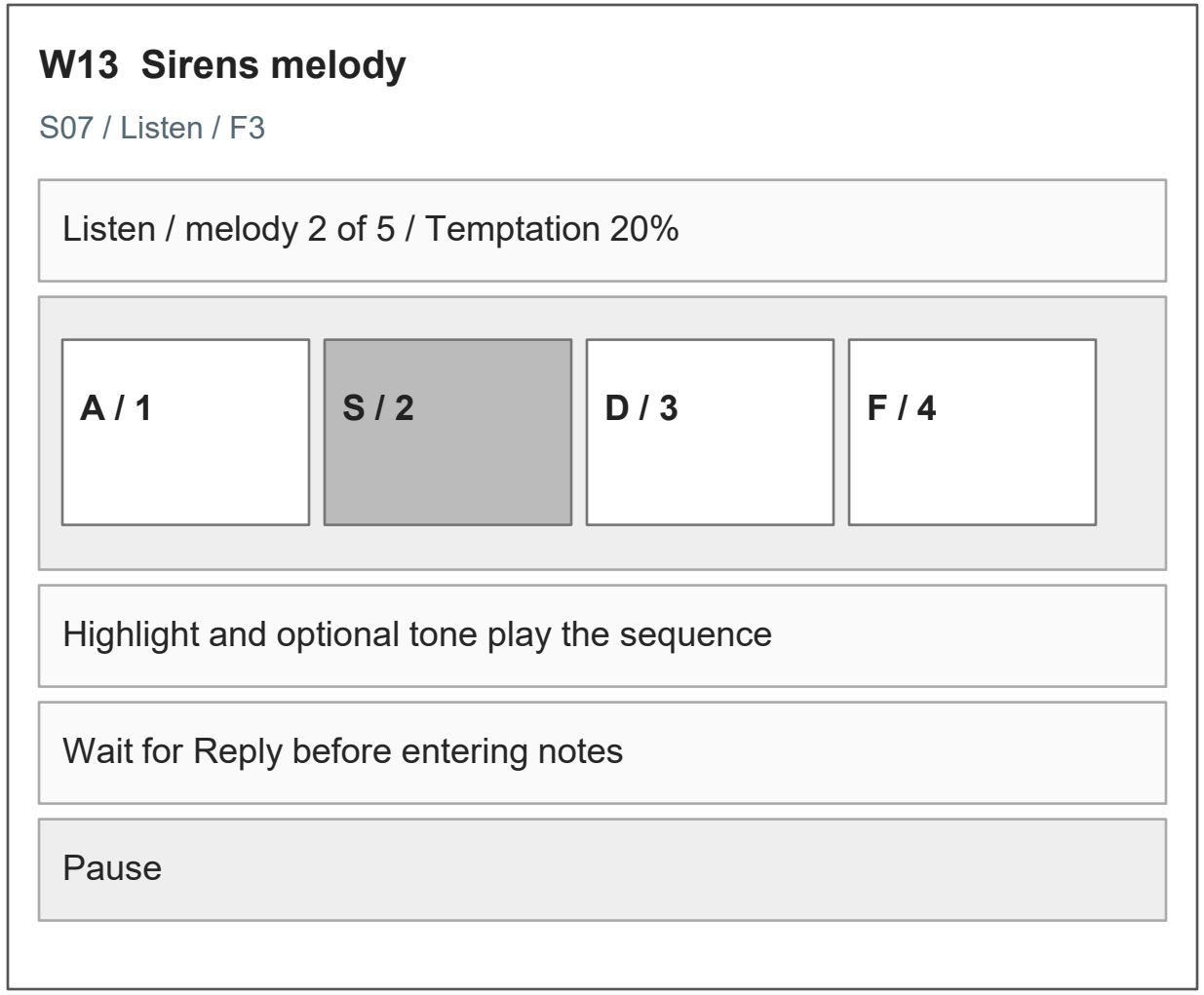
Screen S06

Flow F3

State simultaneous play

# W13: Sirens melody

Review appendix / low-fidelity screen and state




## Implemented

Numbered pads and phase text work without audio. Trick rests appear visibly.

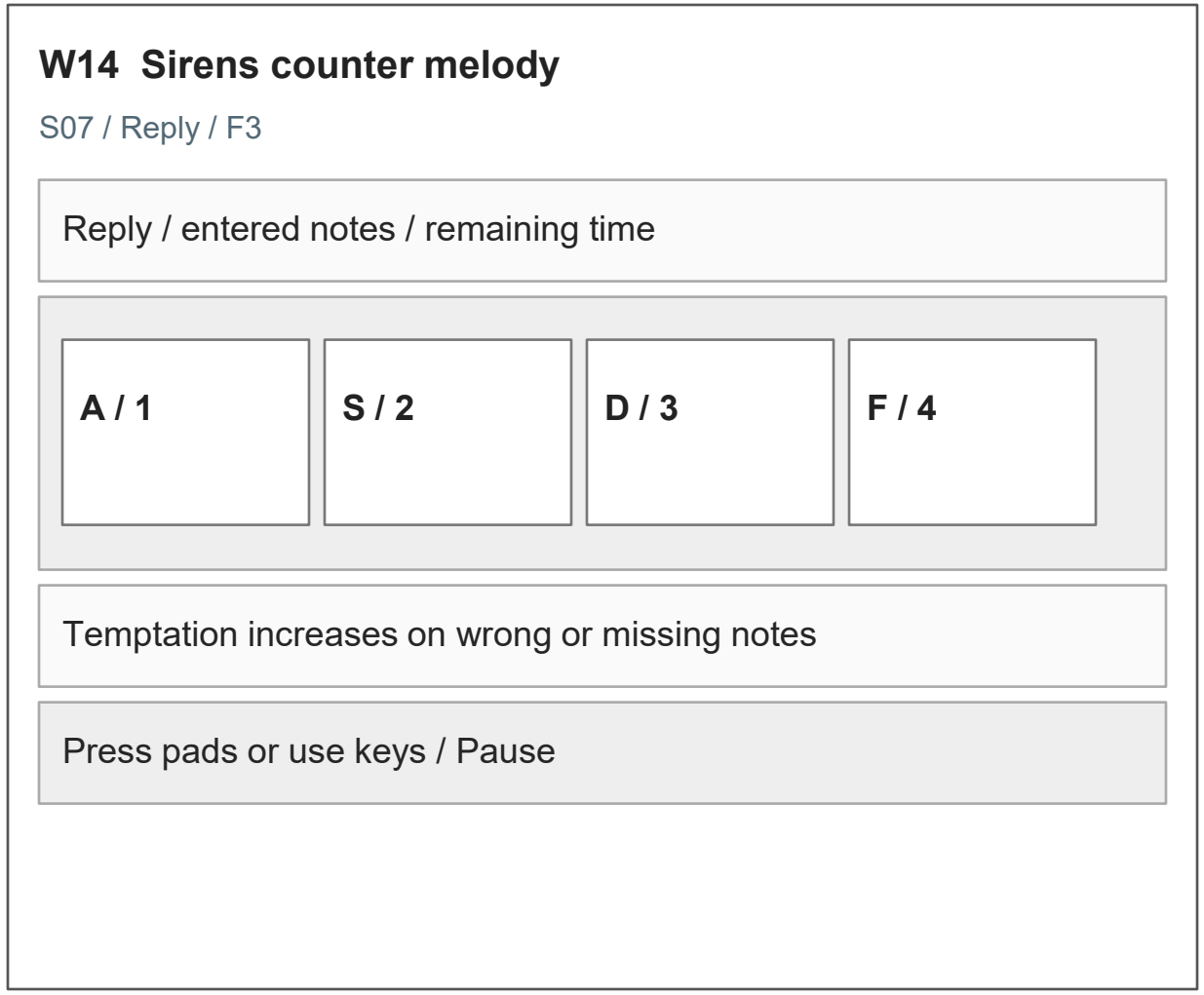
Screen S07

Flow F3

State Listen

# W14: Sirens counter melody

Review appendix / low-fidelity screen and state




## Implemented

Local Siren logic targets misses within the current round. Previous-round misses inform model policy selection, not initial lane targeting. Pads also have labels.

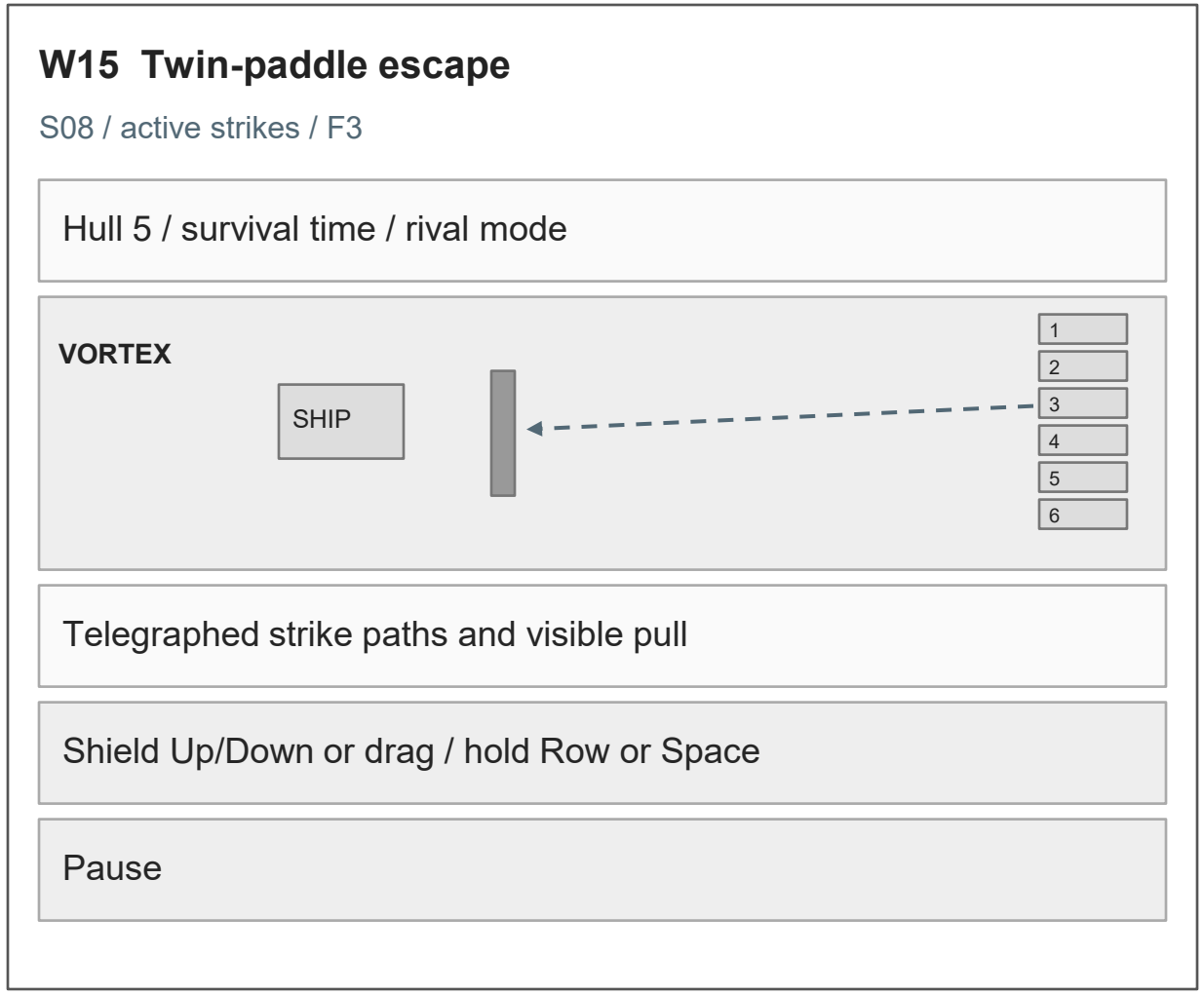
Screen S07

Flow F3

State Reply

# W15: Twin-paddle escape

Review appendix / low-fidelity screen and state




## Implemented

Release Row to stop rowing. Pause or focus loss releases every held input.

Screen S08

Flow F3

State active strikes

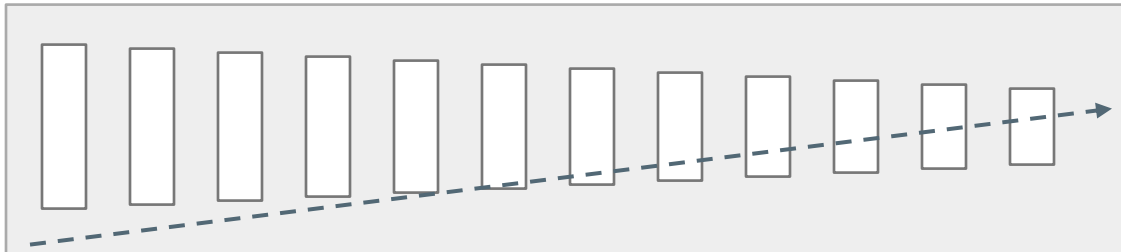
# W16: Archery - player turn

Review appendix / low-fidelity screen and state

**W16  Archery - player turn**

S09 / aim and release / F3

Your turn / shot 1 of 5 / both ring totals



Wind direction/speed and predicted arrow path

Aim / elevation / tension sliders

Release arrow / Pause

## Implemented

Prediction uses the same physics as the shot. Inputs identify their values and can use keyboard controls.

Screen S09

Flow F3

State aim and release

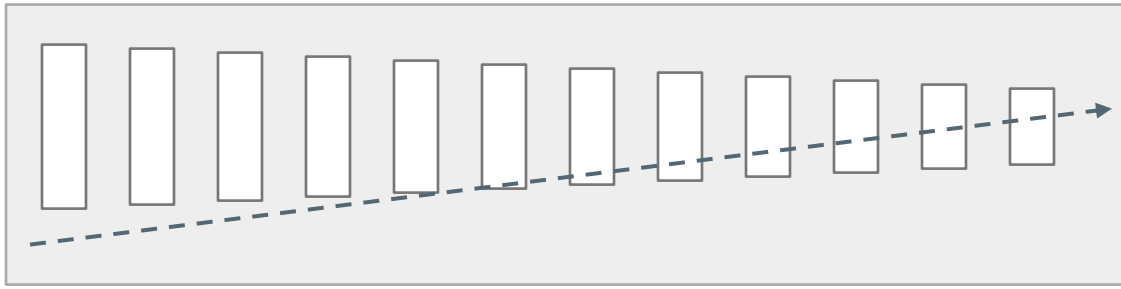
# W17: Archery - rival turn

Review appendix / low-fidelity screen and state

**W17  Archery - rival turn**

S09 / opponent shot / F3

Antinous' turn / shooting controls disabled



Wind Reader or temporary Block-shot action

Next wind appears when player turn resumes

Pause

## Implemented

The rival takes an independent scored turn. A temporary block is visible and expires after one player shot.

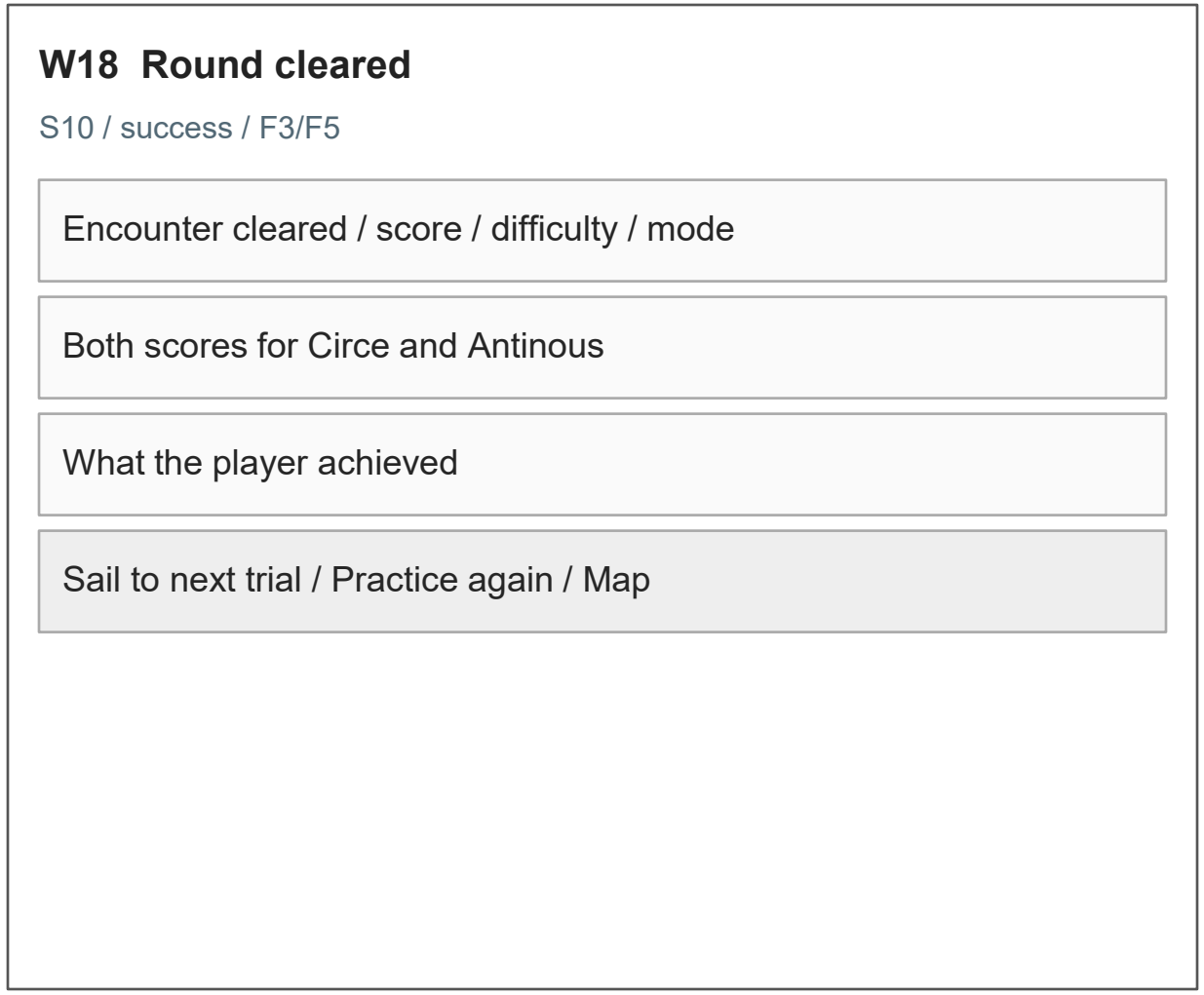
Screen S09

Flow F3

State opponent shot

# W18: Round cleared

Review appendix / low-fidelity screen and state




## Implemented

Update progress once after a full completed round. The final bow clear goes to Homecoming.

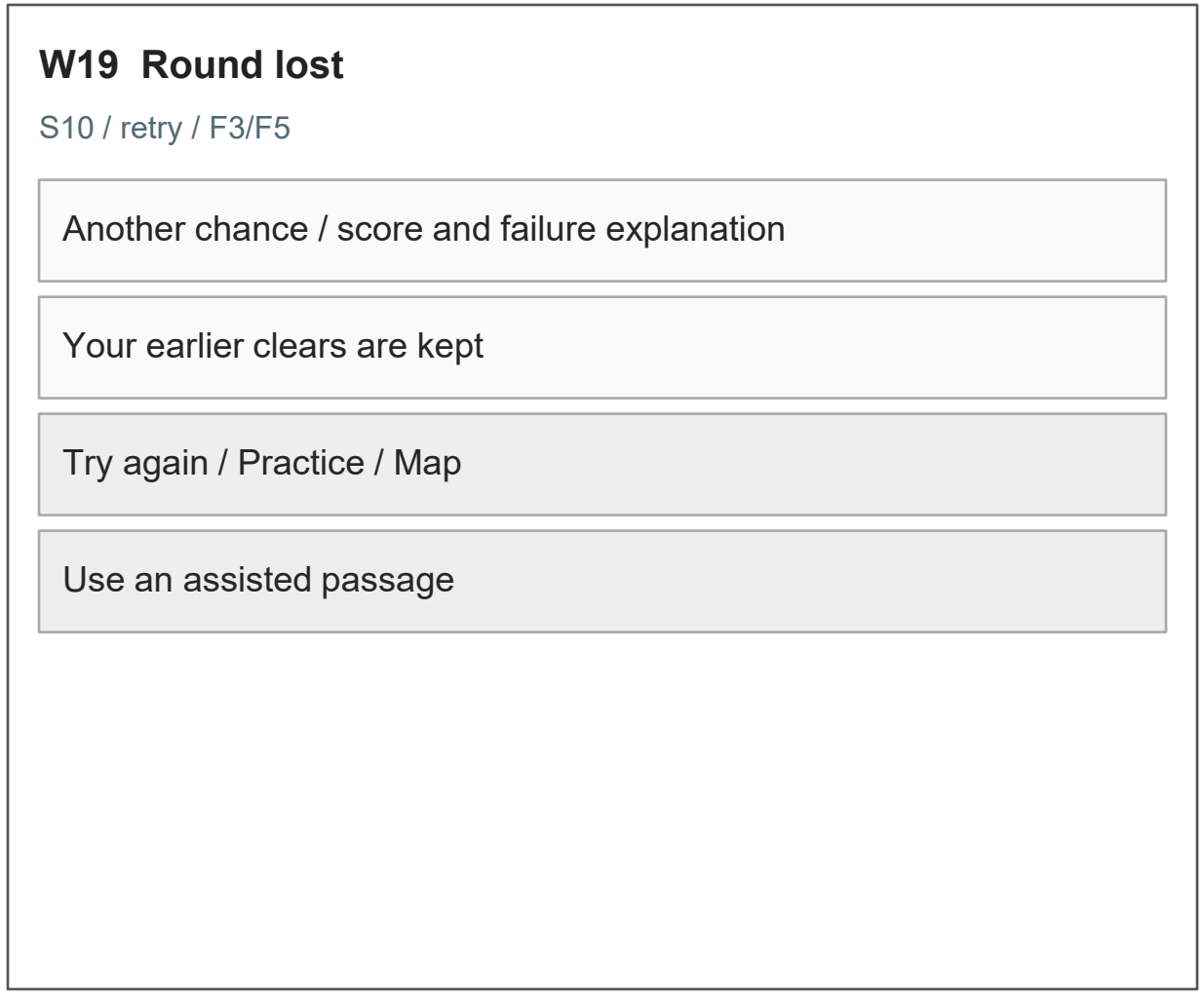
Screen S10

Flow F3/F5

State success

# W19: Round lost

Review appendix / low-fidelity screen and state




## Implemented

Loss keeps a usable path forward. Assisted passage requires S15 confirmation.

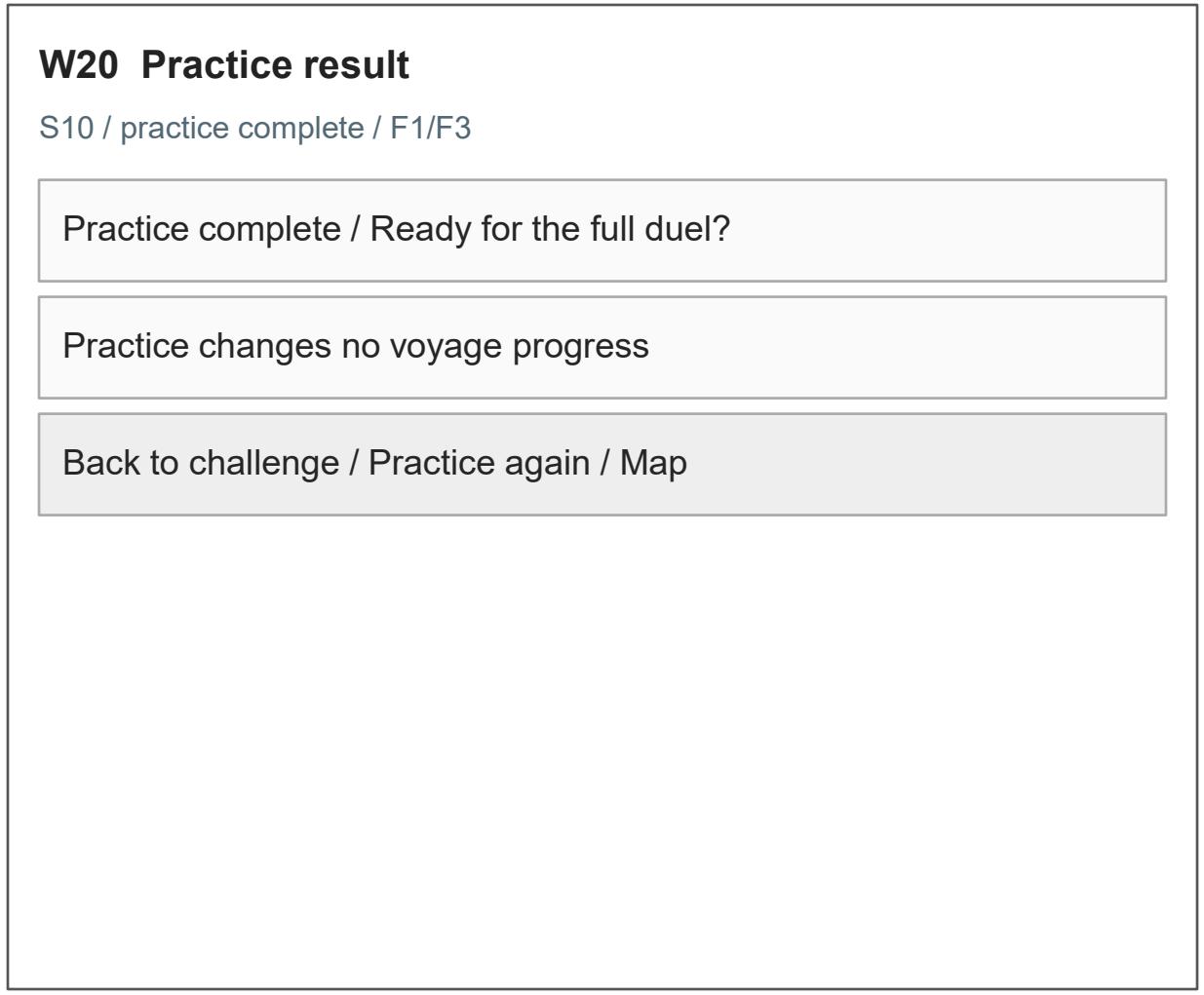
Screen S10

Flow F3/F5

State retry

# W20: Practice result

Review appendix / low-fidelity screen and state




## Implemented

Practice has a separate result label and awards no campaign clear.

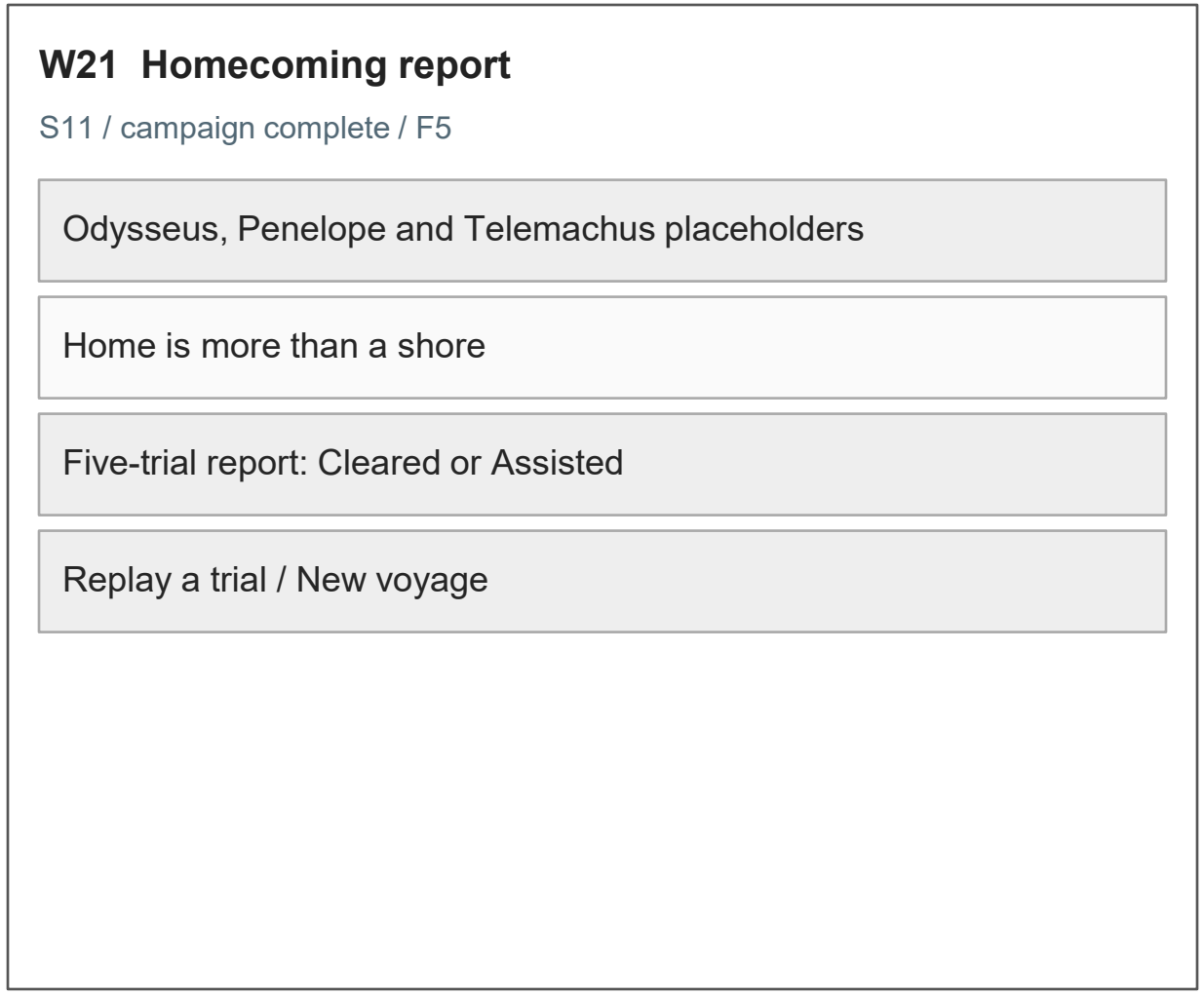
Screen S10

Flow F1/F3

State practice complete

# W21: Homecoming report

Review appendix / low-fidelity screen and state




## Implemented

The ending is authored. The report shows each shore's current Cleared or Assisted status.

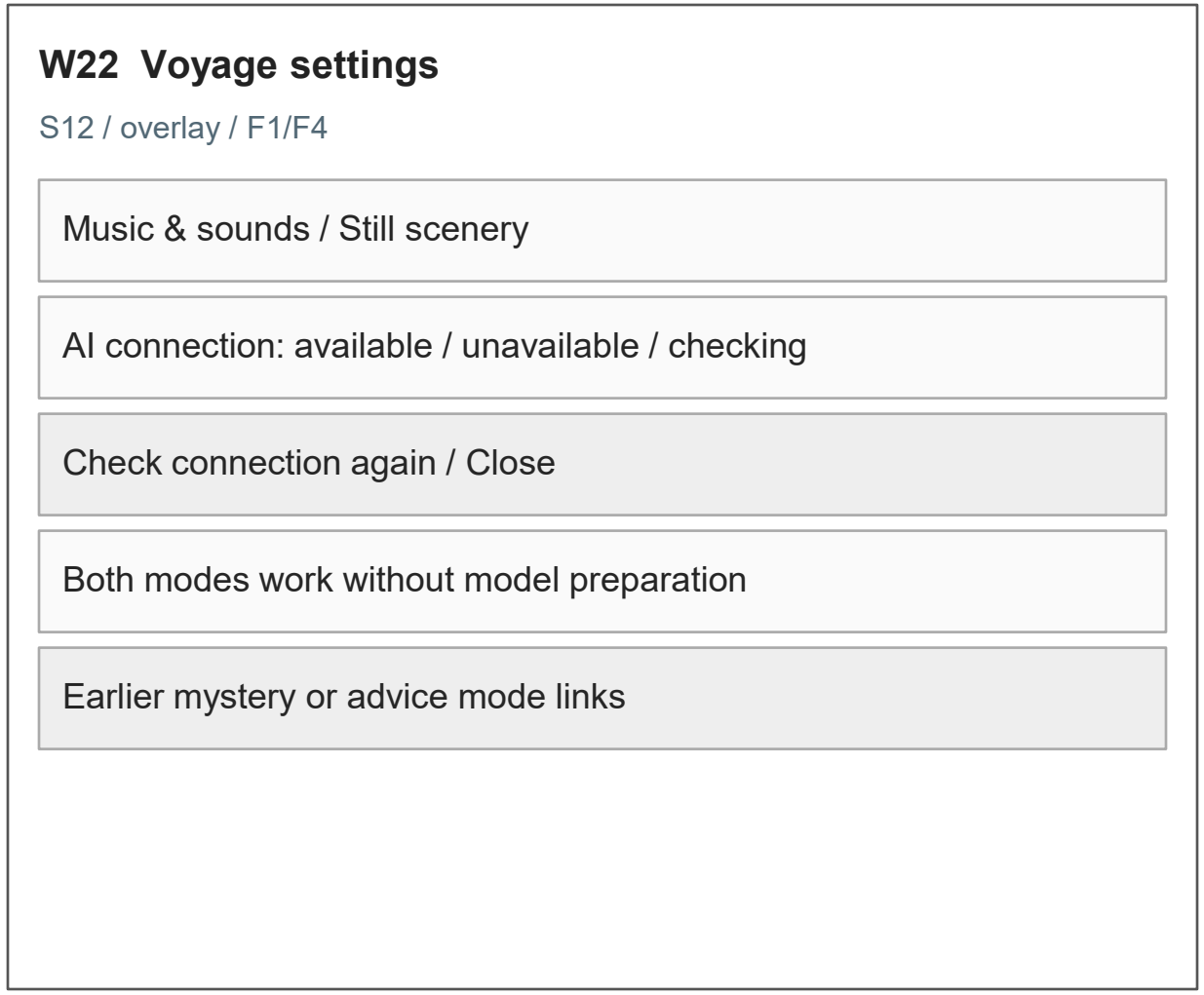
Screen S11

Flow F5

State campaign complete

# W22: Voyage settings

Review appendix / low-fidelity screen and state




## Implemented

A catalog check is separate from generating a strategy. Closing returns focus to the main heading.

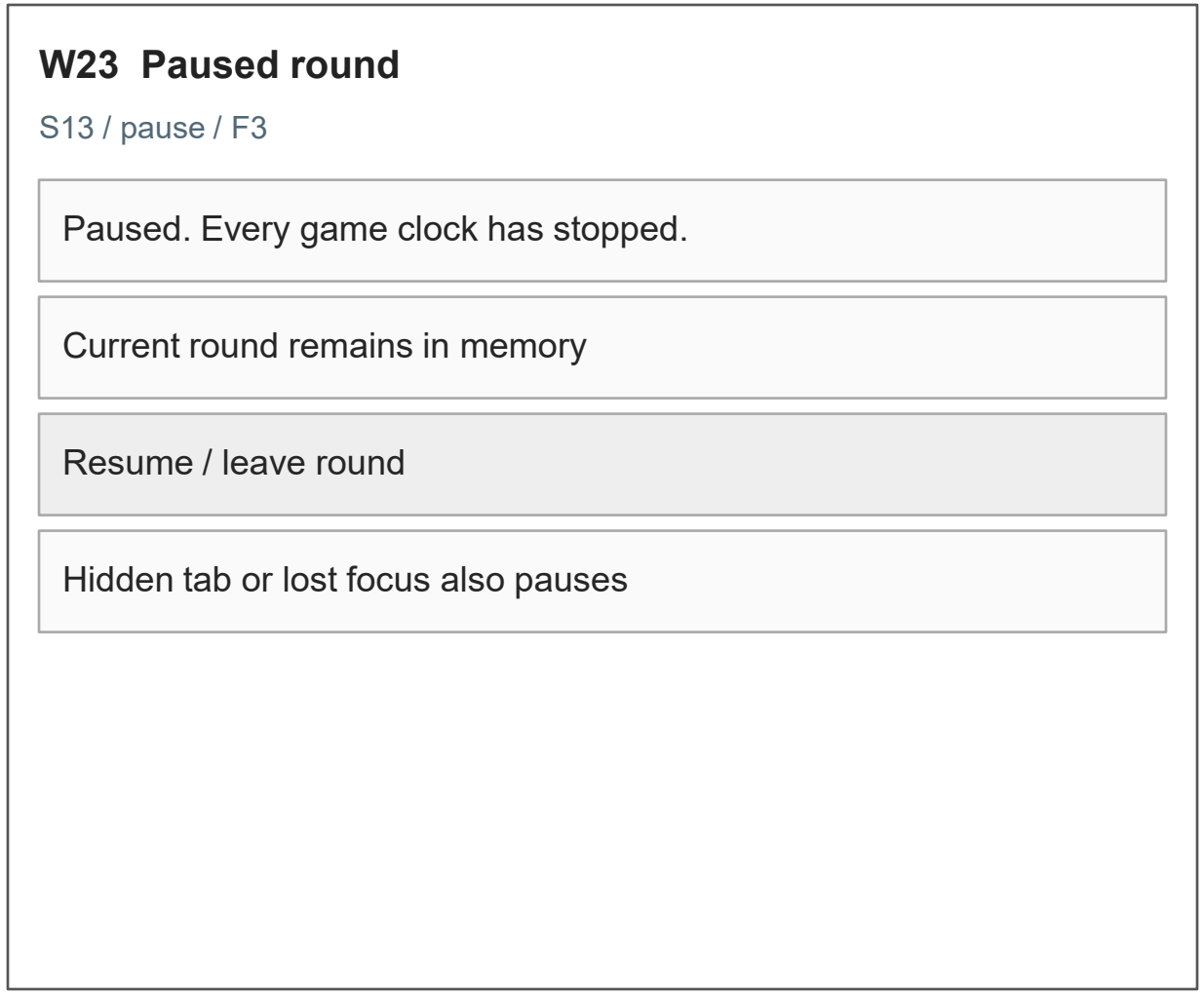
Screen S12

Flow F1/F4

State overlay

# W23: Paused round

Review appendix / low-fidelity screen and state




## Implemented

Resuming does not replay held inputs. The round is not saved across browser reloads.

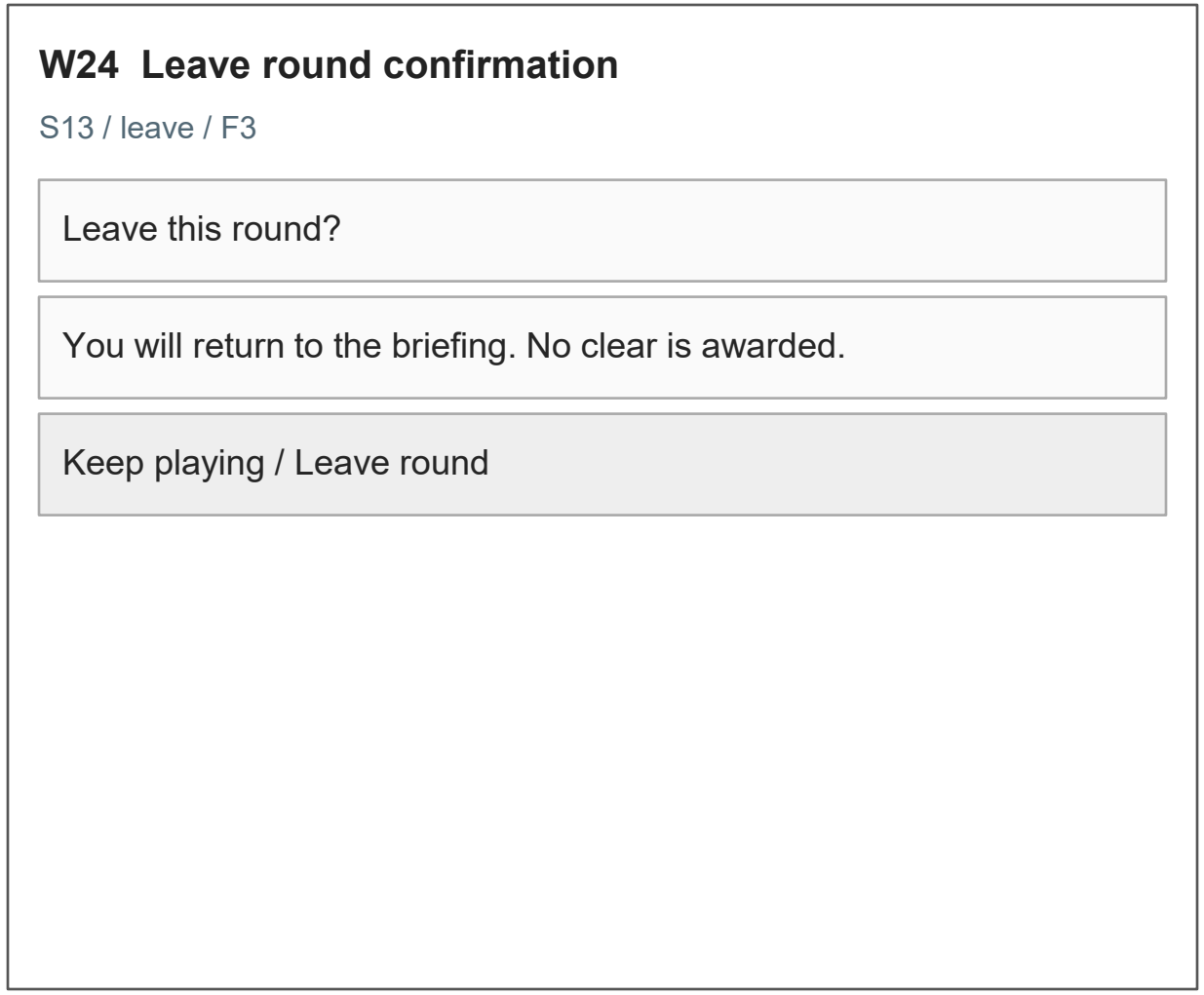
Screen S13

Flow F3

State pause

# W24: Leave round confirmation

Review appendix / low-fidelity screen and state




## Implemented

Protect against accidental loss of an in-progress attempt. Completed voyage progress remains.

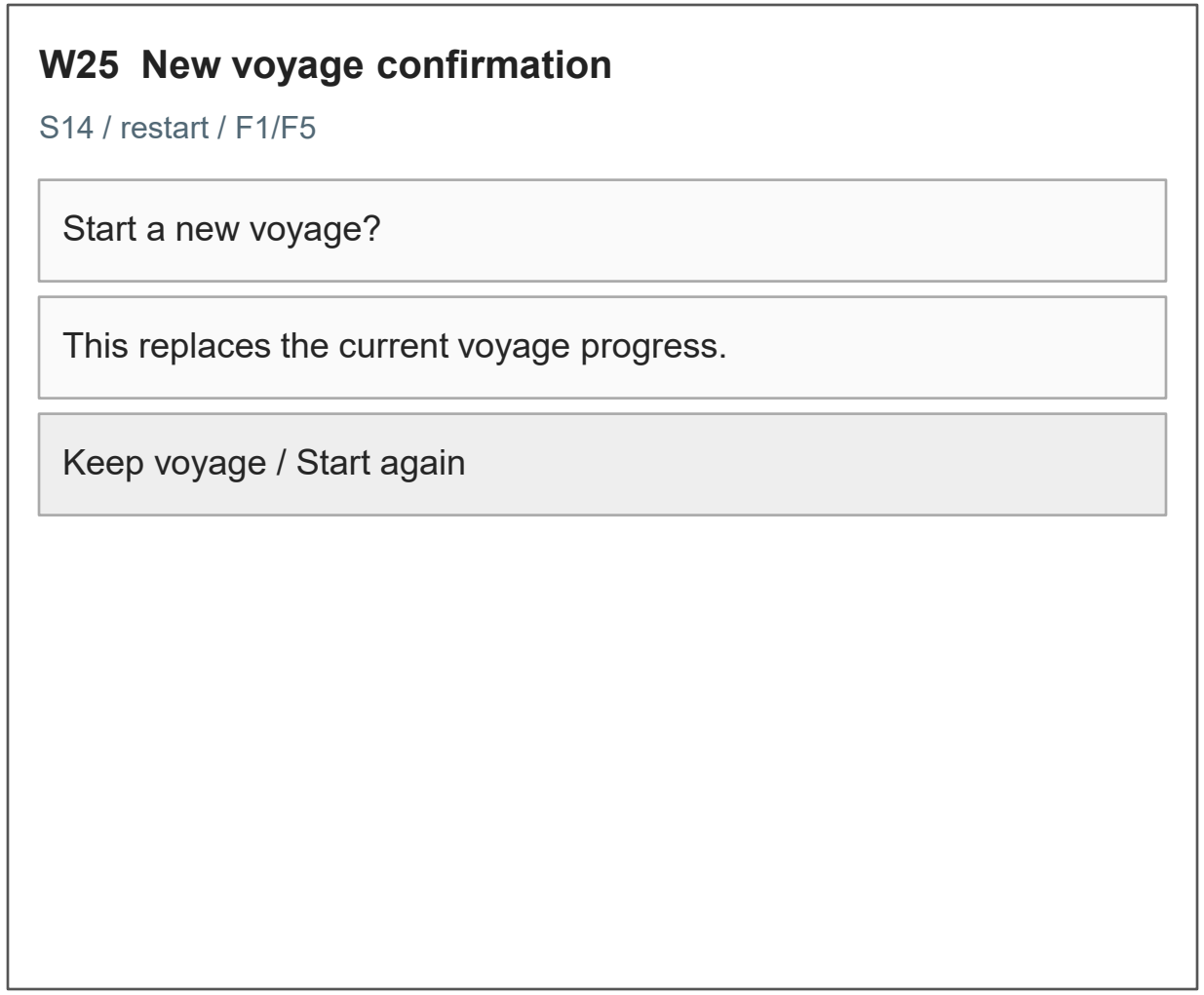
Screen S13

Flow F3

State leave

# W25: New voyage confirmation

Review appendix / low-fidelity screen and state




## Implemented

Only Start again replaces the save. Keep voyage returns unchanged.

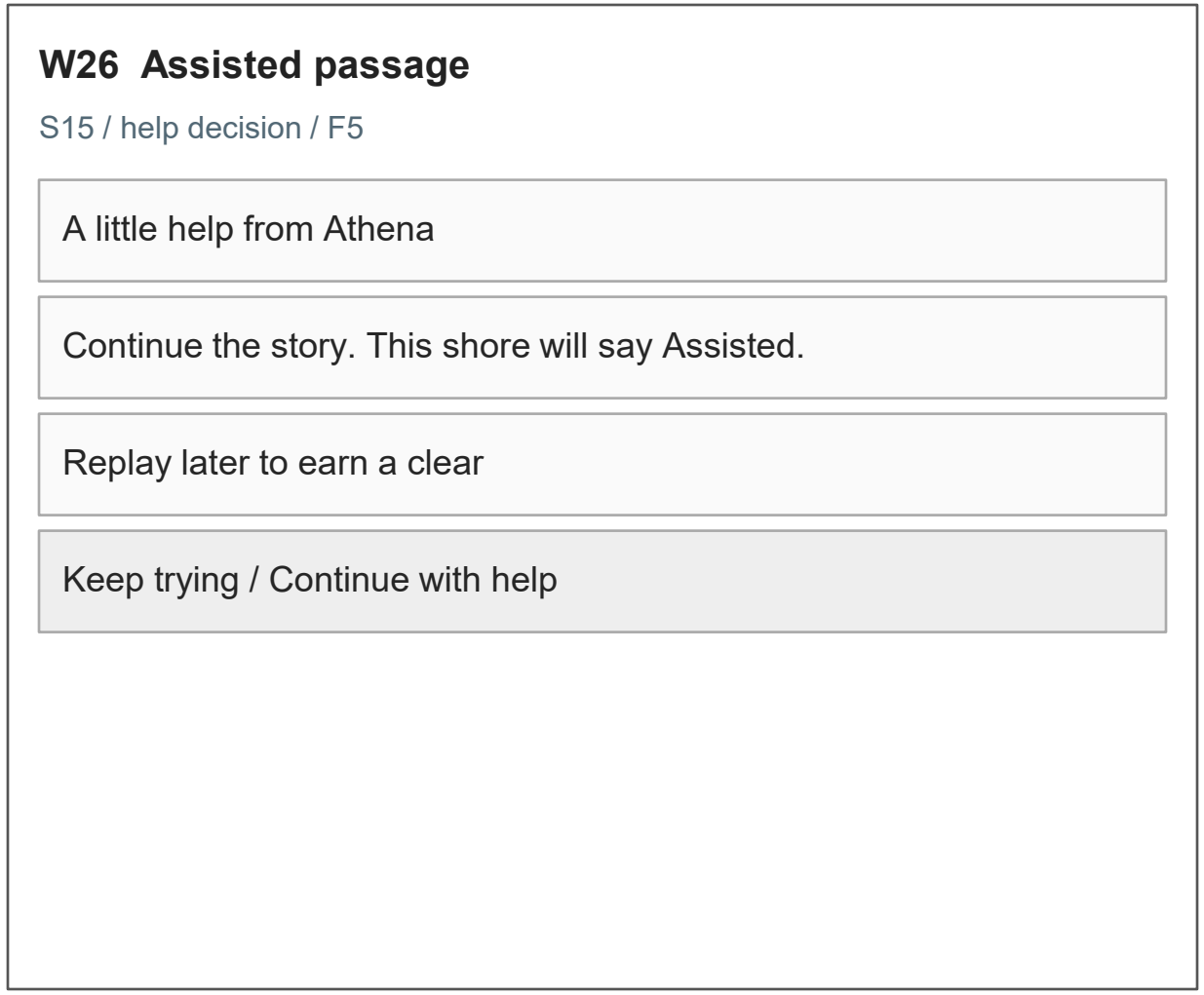
Screen S14

Flow F1/F5

State restart

# W26: Assisted passage

Review appendix / low-fidelity screen and state




## Implemented

Assistance describes exactly how the homecoming report changes.

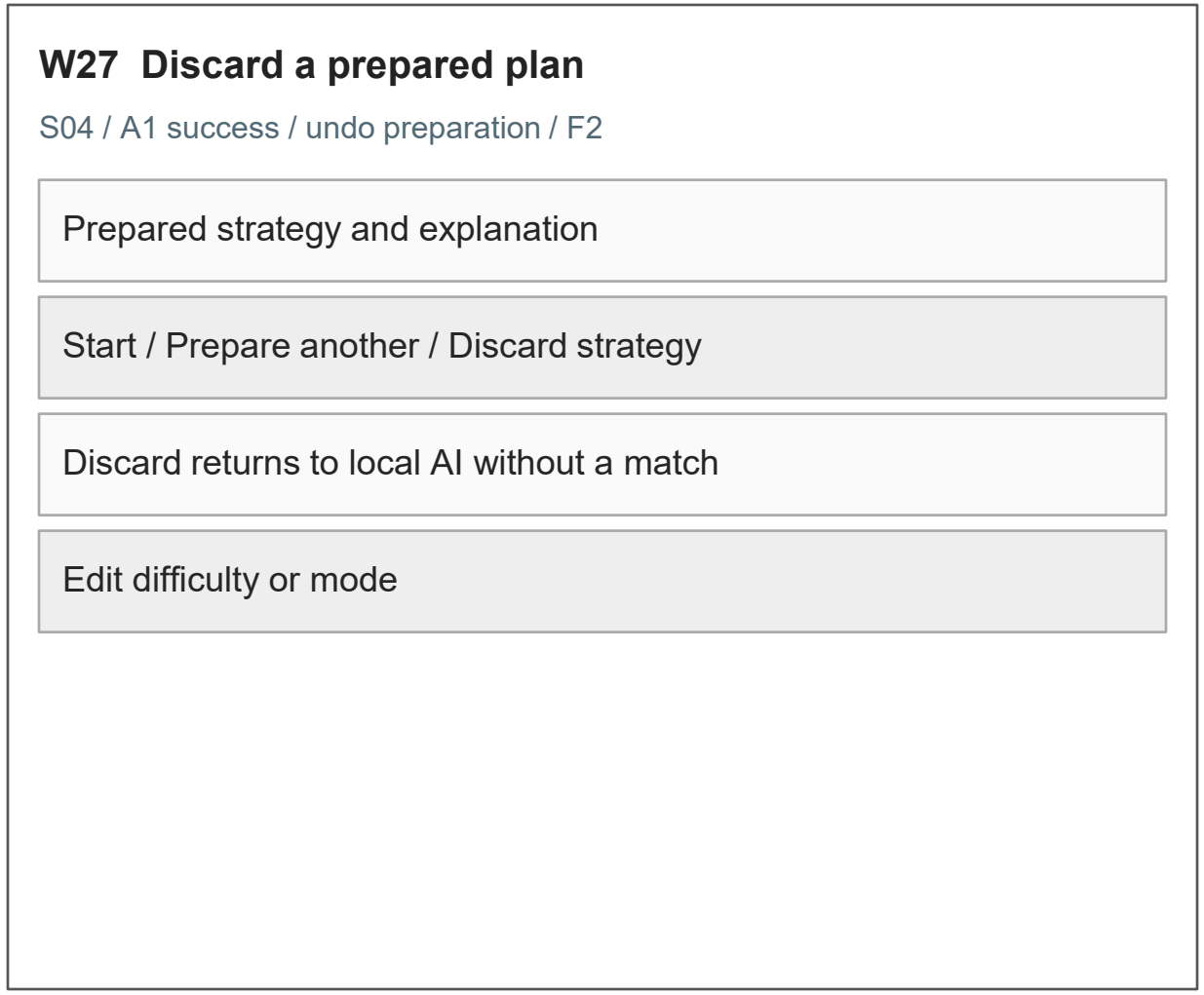
Screen S15

Flow F5

State help decision

# W27: Discard a prepared plan

Review appendix / low-fidelity screen and state




## Proposed

Proposed Discard is an explicit undo of preparation. It clears the unplayed plan. Current edit/mode controls already clear it.

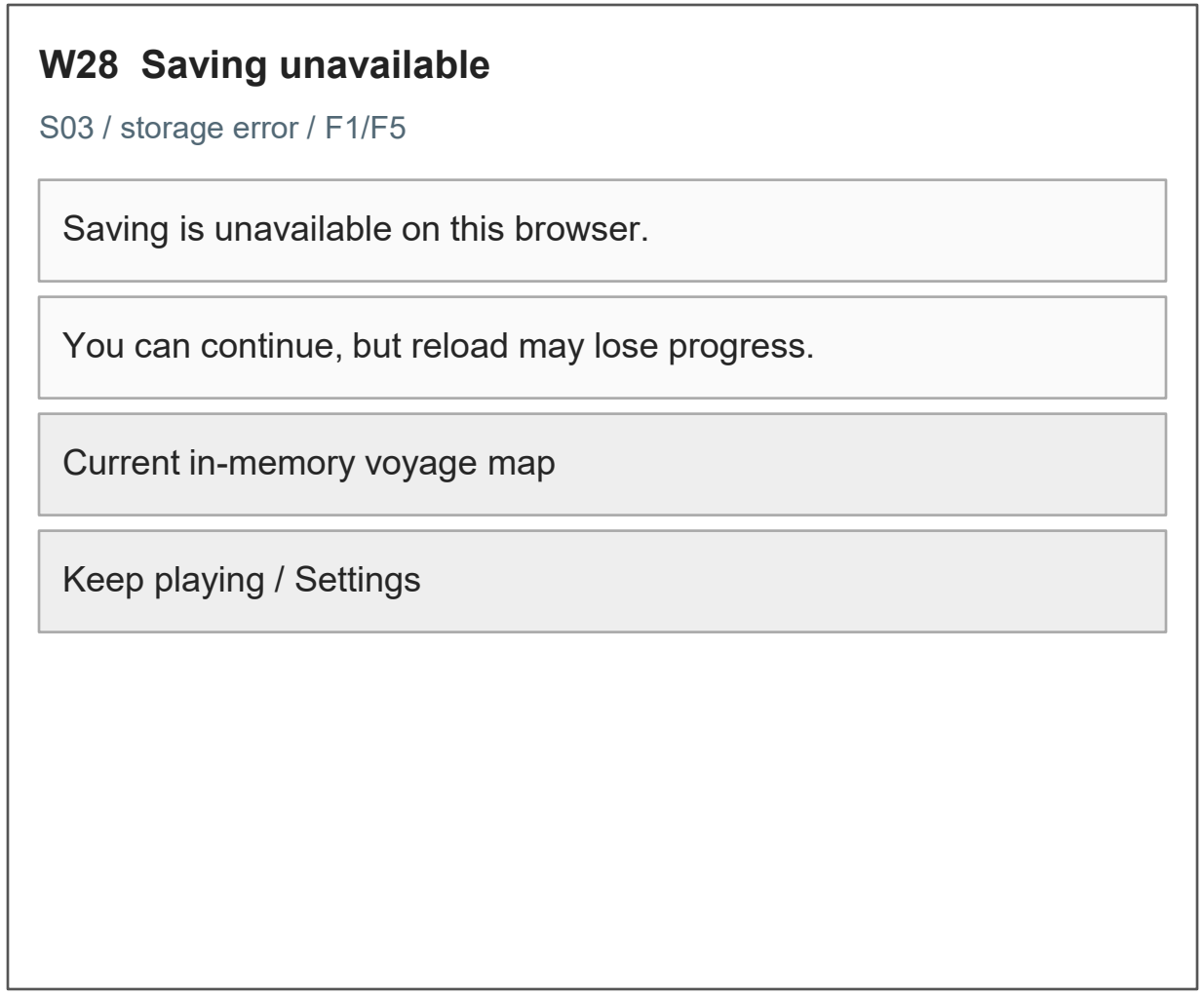
Screen S04

Flow F2

State A1 success / undo preparation

# W28: Saving unavailable

Review appendix / low-fidelity screen and state




## Implemented

A persistent warning makes local storage failure visible while keeping play available.

Screen S03

Flow F1/F5

State storage error

# Exact failure responses

Review appendix / two actual provider failures and one app validation failure

## E1 Nonexistent provider model / HTTP 404

{"error":{"message":"model 'farshore-phase2-deliberately-missing' not found","type":"not_found_error","param":null,"code":null}}

## E2 Malformed provider JSON / HTTP 400

{"error":{"message":"unexpected EOF","type":"invalid_request_error","param":null,"code":null}}

## E3 Unsupported difficulty / HTTP 400

{"error":"Choose a supported encounter and difficulty. At most two attempt summaries are accepted."}

E1 and E2 contact Ollama directly. E3 stops before model inference.